

FC7300F8MDQ Data Sheet

Rev. 0.4

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Chapter 1 Introduction

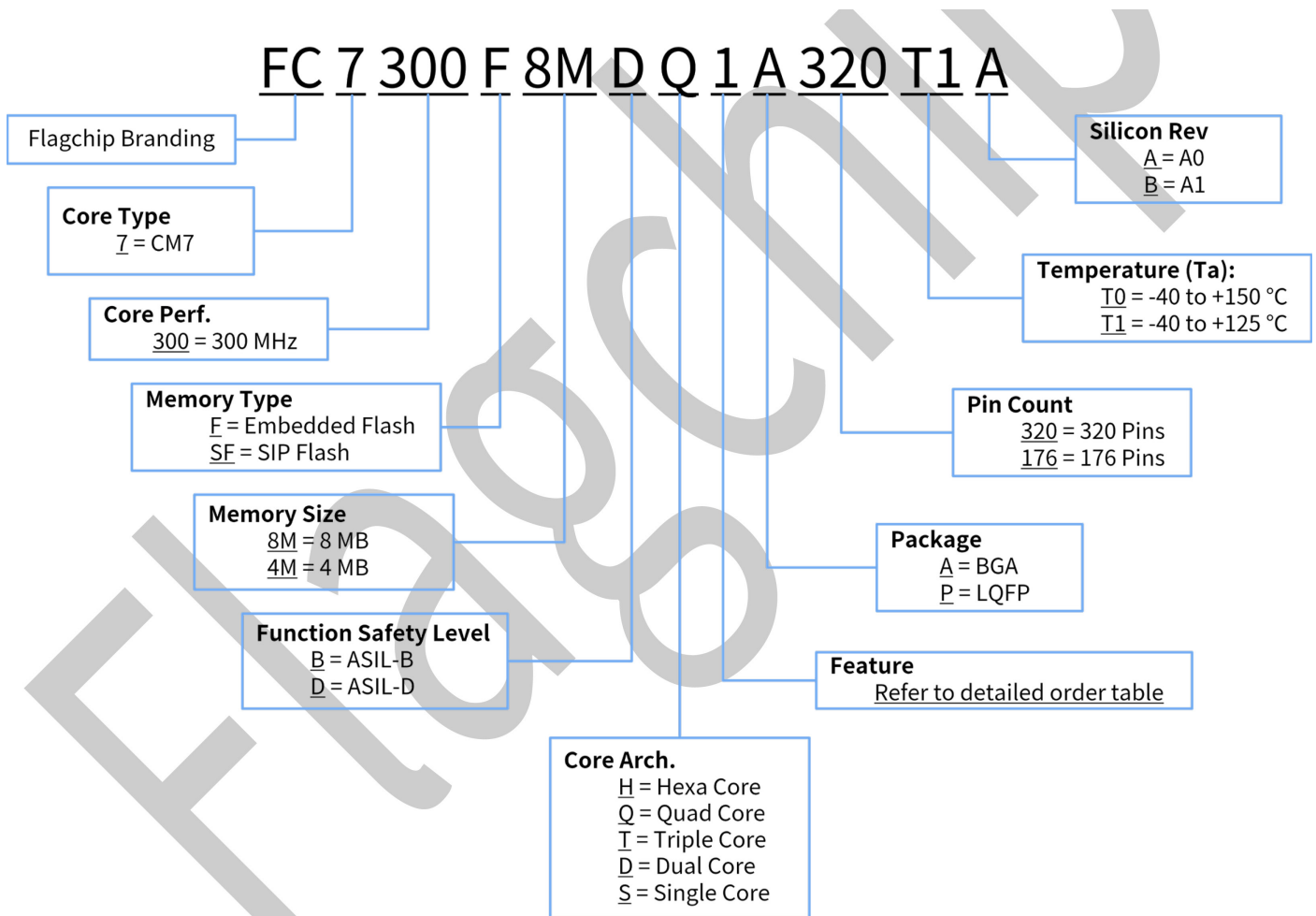
This data sheet provides the ordering information, electrical specifications, package information, and pinout data of the Flagchip FC7300F8MDQ microcontroller (MCU).

NOTE: The specific information of the FC7300F8MDQ family is preliminary until the devices are qualified and may be changed without notice.

1.1 Part Ordering

The ordering of the Flagchip MCU follows the rules below. For detailed part information, refer to the Flagchip company.

Figure 1. Ordering information



NOTE: Not all part number combinations are available.

1.1.1 Orderable Part Number

Refer to the attached *FC7300F8MDQ_Family_Orderable_Part_Number_List.xlsx* for a list of standard orderable part numbers.

1.2 Abbreviations

The following abbreviations are used in this document.

No.	Abbreviation	Description
1	ADC	Analog-to-Digital Converter
2	AFCB	Advanced Flagchip Bus (APB Bridge)
3	AHB	Advanced High-Performance Bus
4	AHBP	AHB-Lite Peripheral
5	AHBS	AHB-Lite Slave
6	AONTIMER	Always-on Timer
7	AP	Access Port
8	APB	Advanced Peripheral Bus
9	AXI	Advanced eXtensible Interface
10	BGA	Ball Grid Array
11	CGC	Clock Gating Control
12	CMU	Clock Monitor Unit
13	CORDIC	Coordinate Rotation Digital Computer
14	CPOL	Clock Polarity
15	CPHA	Clock Phase
16	CRC	Cyclic Redundancy Check
17	CSC	CPU System Control
18	CTI	Cross Trigger Interface
19	CTR	Counter
20	DAC	Digital-to-Analog Converter
21	DDR	Double Data Rate
22	DMA	Direct Memory Access
23	DP	Debug Port
24	DSP	Digital Signal Processing
25	DTCM	Data Tightly-Coupled Memory
26	DWT	Data Watchpoint and Trace
27	ECC	Error Correction Code
28	eFTU	Enhanced Flexible Timer Unit
29	EIM	Error Injection Module
30	ERM	Error Reporting Module
31	ESD	Electrostatic Discharge
32	ETB	Embedded Trace Buffer
33	ETM	Embedded Trace Macro
34	EVITA	E-safety Vehicle Intrusion Protected Applications
35	EVTG	Event Generator
36	FCIIC	Flagchip (FC) Inter-Integrated Circuit
37	FCPIT	Flagchip (FC) Programmable Interrupt Timer
38	FCSMU	Fault Control and Safety Management Unit
39	FCSPI	Flagchip (FC) Serial Peripheral Interface

No.	Abbreviation	Description
40	FCUART	Flagchip (FC) Universal Asynchronous Receiver/Transmitter
41	FF	Flip-Flop
42	FIFO	First In First Out
43	FIRC	Fast Internal Reference Clock
44	FLEXCAN	FLEX Controller Area Network
45	FMC	Flash Memory Controller
46	FOSC	Fast Oscillator
47	FPB	Flash Patch and Breakpoint
48	FPM	Full Performance Mode
49	FPU	Floating Point Unit
50	FREQM	Frequency Measurement
51	FSM	Finite State Machine
52	FTU	Flexible Timer Unit
53	GPIO	General-Purpose Input/Output
54	HMI	Human-Machine Interface
55	HSADC	High Speed Analog-to-Digital Converter
56	HSM	Hardware Secure Module
57	HRPWM	High-resolution PWM
58	HVD	High-Voltage Detect
59	I2S	Inter-IC Sound
60	IIC/I2C	Inter-Integrated Circuit
61	INTM	Interrupt Monitor
62	IRC	Internal Reference Cock
63	ISM	Interface Safety Monitor
64	ISP	In-System Program
65	ITCM	Instruction Tightly-Coupled Memory
66	ITM	Instrumentation Trace Macrocell
67	LBIST	Logic Built-In Self-Test
68	LDO	Low Dropout
69	LIN	Local Interconnect Network
70	LQFP	Low-profile Quad Flat Package
71	LU	Logic Unit
72	LVD	Low-Voltage Detect
73	LVR	Low-Voltage Reset
74	MAM	Matrix Access Monitor
75	MAP	Memory Access Protection
76	MB	Mailbox
77	MBIST	Memory Built-In Self-Test
78	MDC	Management Data Clock
79	MDIO	Management Data Input/Output
80	MII	Media Independent Interface

No.	Abbreviation	Description
81	MISO	Master In Slave Out
82	MOSI	Master Out Slave In
83	MPU	Memory Protection Unit
84	MSC	Micro Second Channel
85	NMI	Non-maskable Interrupt
86	NVIC	Nested Vectored Interrupt Controller
87	NVM	Non-volatile Memory
88	OSC	Oscillator
89	PCC	Peripheral Configuration Controller
90	PLL	Phase-Locked Loop
91	PMC	Power Management Controller
92	POR	Power-on Reset
93	PWM	Pulse Width Modulation
94	QDT	Quadrature Decode Timer
95	RGM	Reset Generation Module
96	RMII	Reduced Media Independent Interface
97	RPM	Reduce Power Mode
98	RSA	Rivest–Shamir–Adleman
99	RTC	Real-Time Clock
100	SAR	Successive Approximation
101	SCG	System Clock Generator
102	SCM	System Control Module
103	SDADC	Sigma Delta Analog-to-Digital Converter
104	SDR	Single Data Rate
105	SEC	Security Controller
106	SECDED	Single-bit Error Correction and Double-bit Error Detection
107	SEMA	Semaphore
108	SENT	Single Edge Nibble Transmission
109	SIRC	Slow Internal Reference Clock
110	SMC	System Mode Controller
111	SOSC	Slow Oscillator
112	SPC	Short PWM Code
113	SPI	Serial Peripheral Interface
114	SSI	Speed and Sensor Interface
115	STCU	Safety Test and Control Unit
116	SWD	Serial Wire Debug
117	TAP	Test Access Port
118	TMU	Temperature Monitor Unit
119	TPIU	Trace Port Interface Unit
120	TRGSEL	Trigger Select
121	TSN	Time-Sensitive Networking

No.	Abbreviation	Description
122	TSTMP	Timer Stamp
123	UART	Universal Asynchronous Receiver and Transmitter
124	VCO	Voltage-Controlled Oscillator
125	WDOG	Watchdog
126	WKU	Wake-up Unit

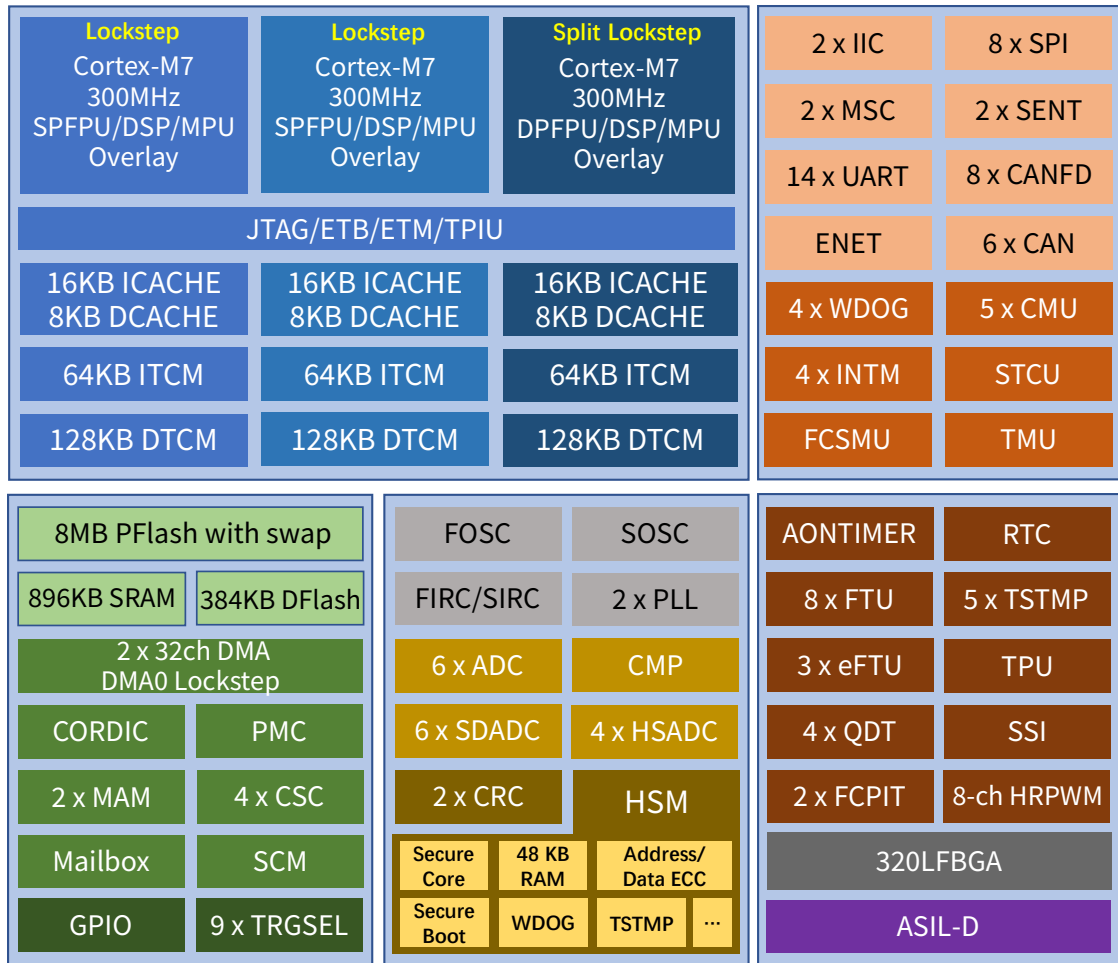
Chapter 2 Features

This chapter summarizes the features of the FC7300F8MDQ series. For detailed information, refer to the *FC7300F8MDQ Reference Manual*. The FC7300F8MDQ product series extends the FC4150 family in the automotive industry with the Arm Cortex-M7 core at higher performance, larger memory and ASILD rating. It also supports security requirements including the secure boot and EVITA full. The FC7300F8MDQ is suitable for a wide range of applications in automotive.

2.1 Block Diagram

The following figures show the block diagrams of the FC7300F8MDQ series.

Figure 2. FC7300F8MDQ block diagram



2.2 Feature Summary

- **Operating Environment**
 - Voltage range: 3.0 V to 5.5 V
 - Support separated 3.0 V and 5.5 V IO rings
 - Ambient temperature range: -40°C to +125°C; junction temperature: -40°C to +150°C
- **Arm Cortex-M7 Core**
 - 300 MHz frequency with 3.23 Dhrystone MIPS per MHz
 - Armv7 Architecture and Thumb-2 ISA
 - Digital Signal Processing (DSP) instruction
 - Single-precision Floating Point Unit (FPU) for CPU0 and CPU1. CPU2 and CPU3 support double precision FPU.
 - Configurable Nested Vectored Interrupt Controller (NVIC)
 - Support Memory Protection Unit (MPU) with 16 regions
 - Each core has 16 KB ICACHE and 8 KB DCACHE
 - Each core has 64 KB ITCM, 64 KB DTCM0 and 64 KB DTCM1. If CPU2 is in split lock mode, each of CPU2/3 has 32KB ITCM, 32KB DTCM0 and 32KB DTCM1.
 - Support mapping TCM to different memory addresses for different CPUs; support configurable SMP and AMP memory maps.
- **Up to four CM7 running cores (CPU0/1/2/3):** Two checker cores to form lockstep for CPU0/1; CPU2 and CPU3 to form lockstep or split lock mode.
- **Support Overlay function,** able to map Flash to system RAM or map a virtual address to Flash or system RAM.
- **CORDIC accelerator for mathematical operations such as angles**
- **Mailbox (MB) with up to 16 communication channels and 5 interrupt channels for on-chip CPU and HSM;** support the hardware semaphore function; support 16 semaphore hardware with resource ID function.
- **Two Matrix Access Monitors (MAMs),** which monitor all on-chip matrix access, and support watchdog timeout function for matrix access.
- **Up to four Core System Control (CSC) modules;** used for the stop mode related handle etc. control.
- **One System Control Module (SCM);** used for the on-chip connection control, lockstep control, CPU low-power status, interrupt routing control, etc.
- **One Security Controller (SEC);** used for on-chip debug/test mode/lifecycle related control.
- **Two 32-channel Direct Memory Access (DMA) modules** with selectable DMA sources. DMA0 with lockstep and DMA1 with address/data transfer monitor.
- **Clock Sources**
 - 16 - 48 MHz Fast Oscillator (FOSC) with up to 50 MHz DC external input clock in bypass mode
 - 32 kHz Slow Oscillator (SOSC)
 - 96 MHz Fast Internal RC Oscillator (FIRC96M)
 - 12 MHz Slow Internal RC Oscillator (SIRC12M)
 - 32 kHz Slow Internal RC Oscillator (SIRC32k)
 - Up to 300 MHz Phased Lock Loop (PLL0 and PLL1) with the reference clock from FIRC48M or FOSC
- **Power Management**
 - Four power modes: RUN, WAIT, STOP and Standby. Optional 64 KB System RAM retention in Standby mode
- **Memory**
 - Up to 8 MB program flash memory with Address and Data Single-bit Error Correction and Double-bit Error Detection (SEDED)

- Up to 384 KB data flash memory with Address and Data SECDED
- Up to 896 KB SRAM and 576 KB TCM (total 1472 KB) with Address and Data SECDED
- 128 KB ROM with Address and Data SECDED, which contains CM7 core self-test/flash program & erase/ Hardware Secure Module (HSM) secure boot etc.
- **Analog**
 - Six 12-bit Successive Approximation (SAR) Analog-to-Digital Converters (ADCs) with up to 24 channel analog inputs for ADC2/3/4/5 and up to 20 channel analog inputs for ADC0/1.
 - ADC1/2 support conversion in Standby mode.
 - Each supports 2-threshold wakeup compare capability.
 - Six Sigma Delta Analog-to-Digital Converters (SDADCs).
 - Support two SDADC wrappers, with each wrapper accommodating three SDADC differential channels.
 - Support the resolver software decode
 - Four 12-bit High Speed Analog-to-Digital Converters (HSADCs). Support the fast compare function
 - One Analog Comparator (CMP) with internal 8-bit Digital-to-Analog Converter (DAC)
- **Debug Functionality**
 - Serial Wire/JTAG Debug Port (SWJ-DP) combines
 - Data Watchpoint and Trace (DWT)
 - Instrumentation Trace Macrocell (ITM)
 - Embedded Trace Macrocell (ETM)
 - Trace Port Interface Unit (TPIU) with up to 16-bit
 - Flash Patch and Breakpoint (FPB) Unit
 - Cross Trigger Interface (CTI)
 - Embedded 4KB Trace Buffer (ETB)
 - FUNNEL
 - JTAG Test Access Port (TAP) and boundary scan support
- **Human-Machine Interface (HMI)**
 - Up to 248 GPIO pins (excl. reset pin) with separated core interrupt support
 - Up to 69 GPIOs with standby wakeup capability
 - Non-Maskable Interrupt (NMI)
 - GPIO interface with separated core control and emergency control
 - Nine Trigger Select (TRGSELS) for on-chip bus connection
 - Two Lookup Unit (LU) modules. Each supports 4 lookup tables and one data unit.
- **Communication Interfaces**
 - 14 FC Universal Asynchronous Receiver/Transmitter (FCUART) modules with LIN support
 - Eight FC Serial Peripheral Interface (FCSPI) modules. Support 1/2/4 data lines and master/slave mode
 - Two FC Inter-Integrated Circuit (FCIIC) modules
 - 14 FLEXCAN modules with optional CAN-FD support for FLEXCAN0-7.
 - Support 96 Message Buffers and 20 Enhanced FIFOs for FLEXCAN0-7.
 - Support 32 Message Buffers for FLEXCAN8-13.
 - One 10/100/1000 Mbps Ethernet with IEEE1588 and TSN, MII/RMII/RGMII support
 - Two Micro Second Channel (MSC) modules
 - Two 4-channel Sent Edge Nibble Transmit (SENT) modules
- **Security**
 - Hardware Secure Module (HSM) with crypto algorithms including AES/SM4/ECC/RAS/SHA/SM3/SM4

- EVITA full capability
- Dedicated Secure Core with dedicated 48 KB RAM, and all 48 KB RAM with address and data ECC protection.
- Optional dedicated 128 KB Data Flash for Secure Core access only
- Optional up to 128 KB SRAM from System SRAM for Secure Core access only
- Support random number generation and pseudo random number generation.
- Key import/export management
- Monotonic counter support
- Support secure boot and non-secure boot mode.
- Support In-System Program (ISP) mode.
- Support FlexCore mode if not required for EVITA full capability.
- **Safety**
 - Five Clock Monitor Units (CMUs) for monitoring all internal critical root clock sources
 - Power Management Controller (PMC) with LVR/LVD/HVD etc., for protection on internally generated supplies and external supplies
 - ECC with both address and data protection on ROM, flash and SRAM memories
 - Memory Access Monitor on the system memory, including peripheral slots such as the APB bridges (AFCB0, AFCB1, and AFCB2)
 - Two Cyclic Redundancy Check (CRC) modules
 - Four internal Watchdogs (WDOGs) with window function
 - One Error Injection Module (EIM) for ECC logic and lockstep logic check
 - One Error Reporting Module (ERM) for ECC-related monitor errors
 - One Fault Control and Safety Management Unit (FCSMU) for on-chip error handling and optionally outputting the chip status to external pins.
 - Lockstep monitor for CPU0/1/2
 - Lockstep monitor for DMA0 and transfer monitor for DMA1
 - End-to-End ECC protection on all masters' access paths
 - One Safety Test Control Unit (STCU) for Logic BIST (LBIST) and Memory BIST (MBIST) controller
 - One Interface Safety Monitor (ISM) module to monitor up to 8 delays/periods/duties of critical signals
 - Four Interrupt Monitor (INTM) modules; each supports up to 4 channels and can monitor the interrupt keep-active timeout or interrupt generation timeout
 - CM7 core self-test API in ROM code
- **Timers**
 - Eight Flexible Timer Unit (FTU) modules with IC/OC/PWM function. FTU1/2 also support the Quadrature Decoder function.
 - One Time Processing Unit (TPU) with up to 32 channels.
 - Support two 24-bit counters TCR1/2. TCR2 also supports angle count mode.
 - Each channel supports 2 matches compare and 2 capture
 - Four Quadrature Decode Timer (QDT) with Home/Index. Support Input Capture function is not used as QD timer
 - Three Enhanced Flexible Timer Units (eFTUs).
 - Each eFTU supports 8 TIMs, 16 TOMs, and 4 DTM Groups.
 - eFTU0 also supports HRPWM output control
 - One High Resolution PWM (HRPWM).
 - Up to 8 channels, and 210ps resolution.
 - Support bypass mode.

- One Speed and Sensor Interface (SSI).
 - Support up to four AK/PWM wheel sensor interfaces.
 - Support PWM Hall Interface
- One Always-on Timer (AONTIMER) with standby wakeup capability
- Four Programmable Timers (PTIMERS) support up to 16 trigger events.
- Two 4-channel FC Programmable Interrupt Timers (FCPITs)
- One Real-Time Clock (RTC)
- Five 56-bit Timer Stamps (TSTMPs) with four 32-bit compare channels. TSTMP0 runs at 1 MHz clock and TSTMP1/2/3/4 run at Bus clock
- One Frequency Measurement (FREQM) module; with up to 64 input clock sources
- **Package**
 - 320LFBGA package

Chapter 3 General

3.1 Absolute Maximum Ratings

The table below lists the maximum allowed conditions for the chip. To avoid the chip damage, the user needs to make sure the conditions are met.

NOTE: VDDA comprises VDDA_0 and VDDA_1, VREFH comprises VREFH_0 and VREFH_1, and VREFL comprises VERFL_0 and VREFL_1.

Table 1. Absolute maximum ratings¹

Symbol	Description	Min.	Max.	Unit
V _{DD_HV_A}	3.0 V - 5.5 V input supply voltage	-0.3	6.0 ²	V
V _{DD_HV_B}	3.0 V - 5.5 V input supply voltage	-0.3	6.0 ²	V
V _{I1}	Core external supply voltage	-0.3	1.45 ³	V
V _{REFH}	3.3 V - 5.0 V high reference voltage	-0.3	6.0 ²	V
V _{REFL}	ADC low reference voltage	-0.3	0.3	V
I _{INJPAD_DC_ABS}	Continuous DC input current (positive / negative) that can be injected into an I/O pin	-	5	mA
V _{IN_DC}	Continuous DC Voltage on any I/O pin with respect to V _{SS}	-0.8	6.0 ²	V
I _{INJSUM_DC_ABS}	Sum of injected currents on all the pins (Continuous DC limit, positive / negative)	-	50	mA
T _{ramp_MCU}	MCU supply ramp rate	-	100 V/ms	-
T _A	Ambient temperature	-40	125	°C
T _{STG}	Storage temperature	-55	165	°C
V _{IN_TRANSIENT}	Transient overshoot voltage allowed on I/O pin beyond V _{IN_DC} limit	-	6.8	V

1. When injecting current, ensure that the injected current does not exceed the chip's self-power consumption (e.g., in standby or stop mode). Otherwise, it may cause damage to the chip.
2. Operation with the 6.0 V maximum is allowed for 10 hours over lifetime.
3. Operation with the 1.45 V maximum is allowed for 7.2 hours over lifetime.

3.2 Operation Condition

The table below lists the chip operation condition. To meet the design specifications, these conditions need to be met.

Table 2. Operating requirements

Symbol	Description	Min.	Typ.	Max.	Unit
V _{DD_HV_A}	Supply voltage	3.0	-	5.5	V
V _{DD_HV_B}	Supply voltage	3.0	-	5.5	V
V _{DD_OFF}	Voltage allowed to be developed on V _{DD_HV_A} pin when it is not powered from any external power supply source.	0	-	0.1	V
V _{DDA}	Analog supply voltage	3.0	-	5.5	V
V _{DD_HV_A} - V _{DDA}	V _{DD_HV_A} to V _{DDA} differential voltage	-0.25	-	0.25	V
V _{DD_HV_A} - V _{DD_HV_B}	V _{DD_HV_A} to V _{DD_HV_B} differential voltage	-0.3	-	-	V

Table 2. Operating requirements (continued)

Symbol	Description	Min.	Typ.	Max.	Unit
V _{REFH}	ADC reference voltage high	3.0	-	V _{DDA} + 0.1	V
V _{REFL}	ADC reference voltage low	-0.1	-	0.1	V
I _{INJPAD_DC_OP}	Continuous DC input current (positive / negative) that can be injected into an I/O pin	-1	-	+1	mA
I _{INJSUM_DC_OP}	Continuous total DC input current that can be injected across all I/O pins	-	-	10	mA
V11	Supply voltage for Core domain	1.12	1.15	1.18	V
V25	Supply voltage for PLL, OSC, FIRC and Flash	2.25	-	2.75	V
T _{pulse}	Reset/NMI input analog filter pulse width	42	-	-	ns

Power Sequencing Requirements**1. External V11 mode:**

- During power-up: VDD_HV_A and VDDA must be powered on simultaneously, and before or at the same time as VREFH. Additionally, VDD_HV_A and VDDA must be powered on before the external V11 supply.
- During power-down: VDD_HV_A and VDDA must be powered off simultaneously, and after or at the same time as VREFH. Additionally, the external V11 supply must be powered off before VDD_HV_A and VDDA.

2. External NMOS mode:

- During power-up: VDD_HV_A and VDDA must be powered on simultaneously, and before or at the same time as VREFH. Additionally, VDD_HV_A and VDDA must be powered on before the V15 supply (for the external NMOS).
- During power-down: VDD_HV_A and VDDA must be powered off simultaneously, and after or at the same time as VREFH. Additionally, the V15 supply (for the external NMOS) must be powered off before VDD_HV_A and VDDA.

3.3 Thermal Operating Condition**Table 3. Thermal operating condition**

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
T _A	Ambient temperature	-40	-	125	°C	
T _J	Junction temperature	-40	-	150	°C	

3.4 Clock Operating Condition

The table below lists the maximum functional clocks of modules. For the IO-related max clock frequency, refer to each IP's electrical specification. The SCG DIVH/M/L outputs should not exceed 160 MHz/ 160 MHz/ 80 MHz respectively.

Table 4. Max. functional clock of modules

Module/Peripheral	Max. Functional Clock (for STA)	Notes
CPU & System Modules		
CM7	300 MHz	
ITCM/DTCM	300 MHz	
ICache and DCache	300 MHz	
CM7 AXI/AHBP/AHBS interface	300 MHz	

Table 4. Max functional clock of modules (continued)

Module/Peripheral	Max. Functional Clock (for STA)	Notes
DMA	150 MHz	
CSCx	150 MHz	
SCM	150 MHz	
WKU	75 MHz	
RGM	75 MHz	
MB	150 MHz	
Memory Modules		
ROM	150 MHz	
System RAM	150 MHz	
FMC	150 MHz	
FC	150 MHz	
Clocking		
SCG	12 MHz	
PCC	75 MHz	
Security Modules		
CRC	150 MHz	
HSM	150 MHz	
Safety Modules		
EIM	75 MHz	
ERM	150 MHz	
INTMx	150 MHz	
ISM	150 MHz	
STCU	75 MHz	
FCSMU	12 MHz	
CMUx	150 MHz	
WDOGx	75 MHz	
Communication Modules		
FCSPiX	160 MHz (pre divider); 80 MHz (after divider)	
FCUARTx	80 MHz	
FCIICx	160 MHz (pre divider); 80 MHz (after divider)	
FLEXCANx	150 MHz	
MSCx	160 MHz (pre/ after divider)	1
SENTx	160 MHz (pre divider); 80 MHz (after divider)	
ENET	25 MHz MII, 50 MHz RMII, 125 MHz RGMII, 2.5 MHz MDC, 150 MHz for PTP clock	
Timer Modules		
AONTIMER0	75 MHz (pre/ after divider)	
FCPITx	160 MHz (pre/ after divider)	
FTUx	150 MHz	

Table 4. Max functional clock of modules (continued)

Module/Peripheral	Max. Functional Clock (for STA)	Notes
eFTUx	150 MHz	
QDTx	150 MHz	
SSI0	80 MHz (pre/ after divider)	
TSTMP0	1 MHz	
TSTMP1-4	150 MHz	
RTC	32 kHz	
PTIMERx	150 MHz	
FREQM	160 MHz	
HMI Modules		
GPIOx	150 MHz	
PORTx	75 MHz	
LUx	75 MHz	
TRGSELx	75 MHz	
Analog Modules		
ADCx	160 MHz (pre divider); 30 MHz (after divider)	
HSADCx	125 MHz (pre divider); 100 MHz (after divider)	
SDADCx	160 MHz (pre divider); 40 MHz (after divider)	
CMP	75 MHz	
PMC	75 MHz	
TMU	75 MHz	

1. The MSC FCLP clock is derived by dividing the functional clock by 2. However, the maximum frequency of the FCLP clock is limited by the pads, so the functional clock cannot be configured to 160 MHz in actual operation.

3.5 LVR, LVD, HVD, and POR Operating Requirements

The chip supports monitors including Power-on Reset (POR)/ Low-Voltage Reset (LVR)/ Low-Voltage Detect (LVD) on the VDD_HV supply.

Table 5. LVR/LVD/HVD and POR

VDD_HV_A/B supply HVD, LVD and POR Operating Ratings						
Symbol	Description	Min.	Typ.	Max.	Unit	Notes
V _{POR}	POR re-arm voltage	1.1	1.6	2.1	V	
V _{LVR_VDD_HV}	LVR on V _{DD_HV_A/B}	2.75	2.85	2.95	V	
V _{HVD_VDD_HV}	HVD on V _{DD_HV_A/B}	5.7	5.85	6.0	V	
V _{LVD_VDD_HV}	LVD on V _{DD_HV_A}	4.3	4.4	4.5	V	
V _{HYS}	Hysteresis Voltage	-	0.04	0.1	V	
V _{LVR_V11}	LVR on V11 and V11 LDO	1.05	-	1.09	V	
V _{HVD_V11}	HVD on V11 and V11 LDO	1.21	-	1.25	V	
V _{BG}	Bandgap voltage reference voltage	1.164	1.2	1.236	V	

3.6 Power Mode Transition

The table below lists the different power mode transition time.

Table 6. Power mode transition time

Symbol	Description	Min.	Typ.	Max.	Unit
t_{POR}	After a POR event, the amount of time from the point VDD reaches 2.7 V to execution of the first instruction across the operating temperature range of the chip.	-	<i>TBD</i>	-	μs
t_{STBtoR}	Standby $\rightarrow$ RUN (PD1 powerup)	-	<i>TBD</i>	-	μs
t_{STtoR}	STOP $\rightarrow$ RUN	-	<i>TBD</i>	-	μs
t_{RtoST}	RUN $\rightarrow$ STOP	-	<i>TBD</i>	-	μs
t_{RtoSTB}	RUN $\rightarrow$ Standby	-	<i>TBD</i>	-	μs
t_{REStoR}	Pin reset $\rightarrow$ RUN	-	<i>TBD</i>	-	μs

3.7 Chip IDD

The chip supports four power modes: RUN/WAIT/STOP/Standby. In Standby mode, the V25 Low Dropout (LDO) and SIRC12M/FOSC etc. can be optionally on. Both code RAM and data RAM can optionally retain 32 KB SRAM in Standby mode. Refer to the *FC7300F8MDQ Reference Manual* for detailed settings.

The table below lists the internally maximum power consumption.

Table 7. Internally maximum power consumption

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
V11	Supply current	-	-	<i>TBD</i>	A	
$V_{DDA} + V_{REFH}$	Supply current	-	-	<i>TBD</i>	mA	
$V_{DD_HV_A}$	Supply current	-	-	<i>TBD</i>	mA	Internal, not including I/Os

The table below lists the chip RUN IDD current. The peripheral enabled/disabled here means enabling/disabling the peripherals' clock gating control (CGC).

Table 8. Chip RUN IDD

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
I_{dd_active}	Active mode current - all peripheral disabled, Core0/1/2/3 while1 code from Flash, 300/100/50 MHz for Core/Bus/Slow, Cache enable	-	<i>TBD</i>	-	mA	
	Active mode current - all peripheral enabled, while1 code from Flash, 300/100/50 MHz for Core/Bus/Slow, Cache enable	-	<i>TBD</i>	-	mA	

Table 9. Chip low power IDD

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
I _{dd_stop}	STOP Mode, 25°C	-	TBD	-	mA	Clock gating, no activity
	STOP Mode 125°C	-	TBD	-	mA	Clock gating, no activity
I _{dd_standby}	Standby Mode, all 64K RAM retention, 25°C	-	TBD	-	μA	
	Standby Mode, only 32K RAM retention, 25°C	-	TBD	-	μA	
	Standby Mode, no RAM retention, 25°C	-	TBD	-	μA	
	Standby Mode, no RAM retention, 125°C	-	TBD	-	μA	
	Standby Mode, no RAM retention, SIRC12M Enable, 25°C	-	TBD	-	μA	
	Standby Mode, no RAM retention, SOSOC Enable, 25°C	-	TBD	-	μA	

3.8 PMC Internal LDO

The chip contains one V25 regulator and one V11 RPM LDO controller. Only V25 supports the Full Performance Mode (FPM), and both V25 and V11 LDO support Reduce Power Mode (RPM). External cap needs to be put on the V25/V11 pin to make sure the LDO and controller function works as expected. If the V11 external mode is selected, the V11 controller will disable and external V11 supply will be applied for the chip core supply.

Table 10. V25 LDO specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
V _{DD_HV_A}	V25 input supply voltage	3	-	5.5	V	
V25	V25 regulator output voltage	2.25	-	2.75	V	
C _{out}	External output capacitor	180	220	330	nF	

Table 11. V11 RPM LDO specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
V _{11_RPM}	Internal V11 RPM regulator output voltage	1.06	1.1	1.21	V	
I _{dd_ldo}	V11 controller power consumption	-	600	-	μA	
C _{out}	External output capacitor	-	10	22	μF	

3.9 ESD Ratings

The Electrostatic Discharge (ESD) result follows the industry test standard.

Table 12. ESD ratings

Symbol	Description		Min.	Max.	Unit	Notes
V_{HBM}	Electrostatic discharge voltage, human body model (HBM)		-2000	2000	V	1, 2, 3
V_{CDM}	Electrostatic discharge voltage, charged-device model (CDM)	All pins except the corner pins	-500	500	V	1, 3, 4
		Corner pins only	-750	750	V	
I_{LAT}	Latch-up current at ambient temperature of 125°C		-100	100	mA	1, 3, 5

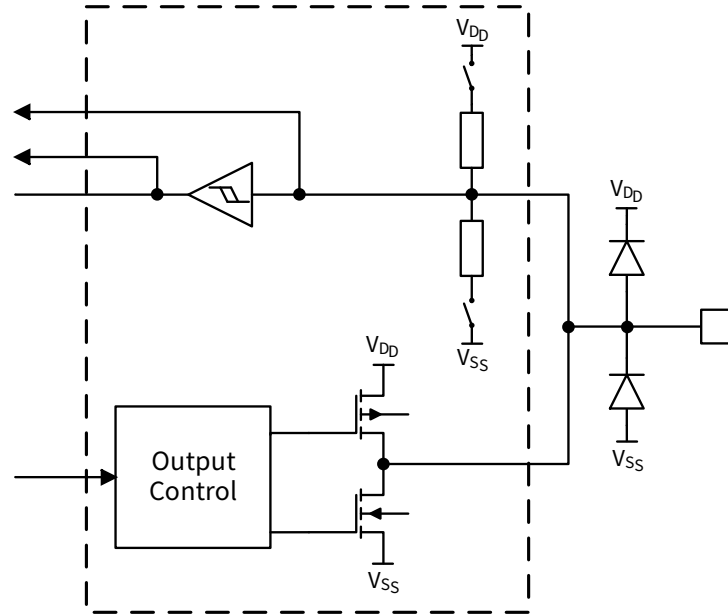
1. Device failure is defined as the situation where the device fails to meet the specification requirements after being exposed to ESD pulses.
2. This parameter is tested in compliance with AEC-Q100-002.
3. All ESD tests comply with AEC-Q100 Stress Test Qualification for Automotive Grade Integrated Circuits.
4. This parameter is tested in compliance with AEC-Q100-011.
5. This parameter is tested in compliance with AEC-Q100-004.

Chapter 4 I/O Parameter

4.1 I/O Structure

The figure below shows the I/O structure of the FC7300F8MDQ series.

Figure 3. I/O structure



4.2 IO DC Specification

The IO can work with supplies from 3.0 V to 5.5 V. The tables below list specifications for the supplies.

Table 13. 3V IO DC specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
V_{IH}	Input high voltage	$0.65 \cdot VDD_IO$	-	-	V	
V_{IL}	Input low voltage	-	-	$0.35 \cdot VDD_IO$	V	
V_{HYS}	Input hysteresis	$0.07 \cdot VDD_IO$	-	-	V	0.09 for normal pad
V_{oh_normal}	I/O current source capability with $I_{oh} = 4mA$	$VDD_IO - 0.8$	-	-	V	
V_{ol_normal}	I/O current sink capability with $I_{ol} = 4mA$	-	-	0.82	V	
V_{oh_fast}	I/O current source capability with $I_{oh} = 4mA$ and $DSE = 0$	$VDD_IO - 0.8$	-	-	V	
V_{ol_fast}	I/O current sink capability with $I_{ol} = 4mA$ and $DSE = 0$	-	-	0.82	V	
V_{oh_fast}	I/O current source capability with $I_{oh} = 8mA$ and $DSE = 1$	$VDD_IO - 0.8$	-	-	V	
V_{ol_fast}	I/O current sink capability with $I_{ol} = 8mA$ and $DSE = 1$	-	-	0.8	V	
V_{oh_ufast}	I/O current source capability with $I_{oh} = 4mA$ and $DSE = 00$	$VDD_IO - 0.8$	-	-	V	

Table 13. 3V IO DC specification (continued)

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
V_{OL_ufast}	I/O current sink capability with $I_{OL} = 4mA$ and DSE = 00	-	-	0.8	V	
V_{OH_ufast}	I/O current source capability with $I_{OH} = 8mA$ and DSE = 01	$VDD_IO - 0.8$	-	-	V	
V_{OL_ufast}	I/O current sink capability with $I_{OL} = 8mA$ and DSE = 01	-	-	0.8	V	
V_{OH_ufast}	I/O current source capability with $I_{OH} = 30mA$ and DSE = 11	$VDD_IO - 0.8$	-	-	V	
V_{OL_ufast}	I/O current sink capability with $I_{OL} = 30mA$ and DSE = 11	-	-	0.8	V	
I_{PU}	Internal pullup current ($R_{load} = 0$, output tie to VSS)	60	100	140	μA	
I_{PD}	Internal pulldown current ($R_{load} = 0$, output tie to VDD_IO)	60	100	140	μA	

Table 14. 5V IO DC specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
V_{IH}	Input high voltage	$0.65 * VDD_IO$	-	-	V	
V_{IL}	Input low voltage	-	-	$0.35 * VDD_IO$	V	
V_{HYS}	Input hysteresis	$0.07 * VDD_IO$	-	-	V	0.09 for normal pad
V_{OH_normal}	I/O current source capability with $I_{OH}=4mA$	$VDD_IO - 0.5$	-	-	V	
V_{OL_normal}	I/O current sink capability with $I_{OL}=4mA$	-	-	0.52	V	
V_{OH_fast}	I/O current source capability with $I_{OH}=4mA$ and DSE=0	$VDD_IO - 0.5$	-	-	V	
V_{OL_fast}	I/O current sink capability with $I_{OL}=4mA$ and DSE=0	-	-	0.52	V	
V_{OH_fast}	I/O current source capability with $I_{OH}=8mA$ and DSE=1	$VDD_IO - 0.5$	-	-	V	
V_{OL_fast}	I/O current sink capability with $I_{OL}=8mA$ and DSE=1	-	-	0.5	V	
V_{OH_ufast}	I/O current source capability with $I_{OH}=4mA$ and DSE=00	$VDD_IO - 0.5$	-	-	V	

Table 14. 5V IO DC specification (continued)

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
V _{OL_ufast}	I/O current sink capability with I _{OL} =4mA and DSE=00	-	-	0.5	V	
V _{OH_ufast}	I/O current source capability with I _{OH} =8mA and DSE=01	VDD_IO - 0.5	-	-	V	
V _{OL_ufast}	I/O current sink capability with I _{OL} =8mA and DSE=01	-	-	0.5	V	
V _{OH_ufast}	I/O current source capability with I _{OH} =30mA and DSE=11	VDD_IO - 0.8	-	-	V	
V _{OL_ufast}	I/O current sink capability with I _{OL} =30mA and DSE=11	-	-	0.8	V	
I _{PU}	Internal pullup current (R _{load} =0, output tie to VSS)	100	160	220	μA	
I _{PD}	Internal pulldown current (R _{load} =0, output tie to VDD_IO)	100	160	220	μA	

4.3 IO AC Specification

The maximum frequency of operation varies from different pad types.

Table 15. Maximum frequency of operation of pads

Pad Type	Maximum Frequency of Operation
Normal pad	25 MHz
High-Speed (HS) pad	50 MHz
Ultra-High-Speed (UHS) pad	125 MHz

The below is the IO AC specification, wherein the minimum value is based on 3.6 V/5.5 V/-40°C condition and the maximum value is based on 3.0 V/4.5 V/150°C condition. **NOTE:** The specification is based on simulation data.

Table 16. 3V IO AC specification

Symbol	SRE	DSE	Rise time(ns)		Fall time(ns)		Delay_Fall		Delay_Rise		Capacitance (pF)
			Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	
t _{RFnormal}	0		1.8	6.3	1.8	6.3	3	17	3	17	15,20%-80% r/f, 50%-50% delay
t _{RFnormal}	0		2.6	9.2	2.6	9.2	6	18	6	18	25,20%-80% r/f, 50%-50% delay
t _{RFnormal}	1		2.2	8.5	2.2	8.5	5.5	22	5.5	22	15,20%-80% r/f, 50%-50% delay
t _{RFnormal}	1		3.2	12	3.2	12	6	25	6	25	25,20%-80% r/f, 50%-50% delay
t _{RFfast}		1	1	3.7	1	3.7	3.1	9.5	3.1	9.5	15,20%-80% r/f, 50%-50% delay
t _{RFfast}		1	1.2	4.7	1.2	4.7	3.5	11	3.5	11	25,20%-80% r/f, 50%-50% delay
t _{RFfast}		0	1.7	6.5	1.7	6.5	3.2	9.5	3.2	9.5	15,20%-80% r/f, 50%-50% delay

Table 16. 3V IO AC specification (continued)

Symbol	SRE	DSE	Rise time(ns)		Fall time(ns)		Delay_Fall		Delay_Rise		Capacitance (pF)
			Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	
t _{RFfast}		0	2.7	8.9	2.7	8.9	4.0	12	4.0	12	25,20%-80% r/f, 50%-50% delay
t _{RFufast}		11	0.24	1	0.24	1	2.3	7.8	2.3	7.8	15,20%-80% r/f, 50%-50% delay
t _{RFufast}		11	0.34	1.5	0.34	1.5	2.4	8	2.4	8	25,20%-80% r/f, 50%-50% delay
t _{RFufast}		01	0.4	2.8	0.4	2.8	2.3	8.8	2.3	8.8	15,20%-80% r/f, 50%-50% delay
t _{RFufast}		01	0.9	3.8	0.9	3.8	2.5	9.5	2.5	9.5	25,20%-80% r/f, 50%-50% delay
t _{RFufast}		00	1	5.6	1	5.6	3.1	13	3.1	13	15,20%-80% r/f, 50%-50% delay
t _{RFufast}		00	1.9	7.5	1.9	7.5	3.3	14	3.3	14	25,20%-80% r/f, 50%-50% delay

Table 17. 5V IO AC specification

Symbol	SRE	DSE	Rise time(ns)		Fall time(ns)		Delay_Fall		Delay_Rise		Capacitance (pF)
			Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	
t _{RFnormal}	0		1.4	4.8	1.4	4.8	3	13	3	13	15,20%-80% r/f, 50%-50% delay
t _{RFnormal}	0		2.2	6.8	2.2	6.8	3.6	14	3.6	14	25,20%-80% r/f, 50%-50% delay
t _{RFnormal}	1		1.8	6	1.8	6	6	16	6	16	15,20%-80% r/f, 50%-50% delay
t _{RFnormal}	1		2.5	8	2.5	8	6.4	18	6.4	18	25,20%-80% r/f, 50%-50% delay
t _{RFfast}		1	0.75	2.7	0.75	2.7	2.5	7.5	2.5	7.5	15,20%-80% r/f, 50%-50% delay
t _{RFfast}		1	1	3.8	1	3.8	2.9	8	2.9	8	25,20%-80% r/f, 50%-50% delay
t _{RFfast}		0	1.2	4.3	1.2	4.3	2.6	7.5	2.6	7.5	15,20%-80% r/f, 50%-50% delay
t _{RFfast}		0	2	6.9	2	6.9	3.2	9.3	3.2	9.3	25,20%-80% r/f, 50%-50% delay
t _{RFufast}		11	0.2	0.8	0.2	0.8	2	5.7	2	5.7	15,20%-80% r/f, 50%-50% delay
t _{RFufast}		11	0.25	1	0.25	1	2	6	2	6	25,20%-80% r/f, 50%-50% delay
t _{RFufast}		01	0.3	2	0.3	2	2	6	2	6	15,20%-80% r/f, 50%-50% delay
t _{RFufast}		01	0.5	2.8	0.5	2.8	2	7	2	7	25,20%-80% r/f, 50%-50% delay
t _{RFufast}		00	1	3.8	1	3.8	2.2	9	2.2	9	15,20%-80% r/f, 50%-50% delay
t _{RFufast}		00	1.3	5	1.3	5	2.5	10	2.5	10	25,20%-80% r/f, 50%-50% delay

Chapter 5 Clock Specification

5.1 FOSC Specification

The Fast Oscillator (FOSC) supports the range of 16 - 48 MHz. To avoid the startup noise impact, software should wait enough time before using the clock. The FOSC can keep enabled in Stop mode.

Figure 4. Crystal connection diagram

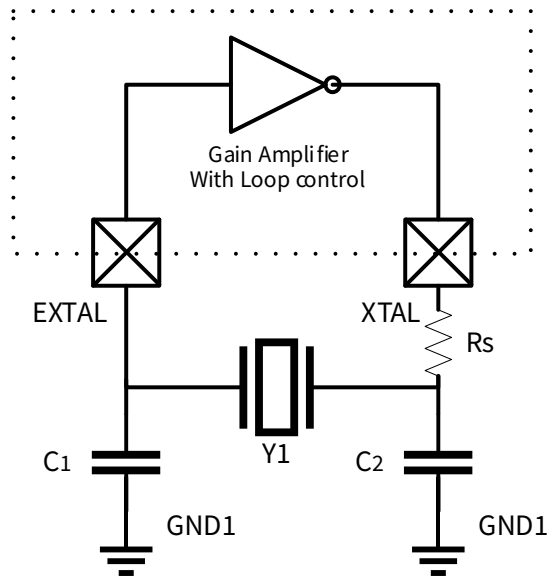


Table 18. FOSC specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
f_{osc}	Oscillator crystal or resonator frequency	16	-	48	MHz	
T_{st_osc}	Startup time	-	600	-	μs	
I_{dd_osc}	Current consumption	-	1200	-	μA	
GM	Gain amplifier transconductance	11.07	-	26.24	mA/V	Select max GM setting

For crystal selection, usually require $g_m > 5 \cdot g_{m_{crit}}$.

- Formula $gain_{margin} = g_m / g_{m_{crit}}$, where:
 - g_m is the oscillator transconductance specified in the data sheet. Note that the oscillator transconductance is in the range of a dozen of mA/V .
 - $g_{m_{crit}}$ is defined as the minimal transconductance of an oscillator required to maintain a stable oscillation when it is a part of the oscillation loop for which this parameter is relevant.
 - $g_{m_{crit}}$ is computed from oscillation-loop passive components parameters (R_s is 0 here).
- $g_{m_{crit}} = 4 \times ESR (2\pi F)^2 \times (C_0 + C_L)^2$, where:
 - ESR is the equivalent series resistance.
 - C_0 is the crystal shunt capacitance.
 - C_L is the crystal nominal load capacitance, $C_L = C_s + [C_1 \cdot C_2 / (C_1 + C_2)]$ (NOTE: C_1 should be equal to C_2).
 - F is the crystal nominal oscillation frequency.

5.2 SOSC Specification

The Slow Oscillator (SOSC) supports the 32.768 kHz crystal. To avoid the startup noise impact, software should wait enough time before using the clock. The SOSC will keep enabled during system reset. Only POR/LVD will reset the SOSC.

Table 19. SOSC (32K) specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
f_{osc32k}	Oscillator crystal or resonator frequency	-	32.768	-	kHz	
T_{st_osc32k}	Startup time	-	1	-	s	
I_{dd_osc32k}	Current consumption	-	1.8	-	μA	
GM	Gain amplifier transconductance	26	-	54.5	$\mu A/V$	

5.3 FIRC96M Specification

The FIRC96M is the default system clock after reset. FIRC96M provides two clock sources, one is 96 MHz and the other is 48 MHz. The 96 MHz clock can be selected as the system clock. The 48 MHz clock can be selected as the PLL reference clock and Flash program/erase clock. Software needs to make sure FIRC is enabled and 48 MHz clock output is enabled before a flash program/erase operation or selecting FIRC as the PLL reference clock. After reset, the FIRC is enabled and both 96 MHz and 48 MHz clock outputs are enabled.

Table 20. FIRC96M specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
$f_{firc96m}$	FIRC96M clock frequency range	-	96	-	MHz	
$\Delta f_{firc96m}$	Frequency deviation with 1T trim	-	-	± 4	$\%f_{firc96m}$	
$T_{st_firc96m}$	Startup time (< 20% accuracy), from disable	-	2.4	-	μs	
	Startup time (< 1% accuracy)	-	40	-	μs	
$I_{dd_firc96m}$	Current consumption (Normal run mode)	-	100	-	μA	
	Current consumption (Disable)	-	100	-	nA	
T_{valid}	Valid assertion time ¹	-	6.5	20	μs	

1. The time is for reference only. It is measured from enable assert to valid bit assert.

5.4 SIRC12M Specification

The SIRC12M is used for system low power mode entry. SIRC12M can keep enabled in Standby mode. In RUN mode, SIRC12M is always enabled. Clock sources of WDOGx/TSTMP0 etc. can be selected from SIRC12M divider outputs.

Table 21. SIRC12M specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
$f_{sirc12m}$	SIRC12M clock frequency	-	12	-	MHz	
$\Delta f_{sirc12m}$	Frequency deviation with 1T trim	-	-	± 5	$\%f_{sirc12m}$	
$T_{st_sirc12m}$	Startup time from POR	-	3	-	μs	
$I_{dd_sirc12m}$	Current consumption	-	8	-	μA	Analog only
	Current consumption (Disable)	-	100	-	nA	
T_{valid}	Valid assertion time ¹	-	30	100	μs	

1. The time is for reference only. It is measured from enable assert to valid bit assert.

5.5 SIRC32k Specification

The SIRC32k can be used for low power wakeup source. It can keep enabled in Standby mode.

Table 22. SIRC32k specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
f_{sirc32k}	SIRC32k clock frequency	-	32	-	kHz	
$\Delta f_{\text{sirc32k}}$	Frequency deviation with 1T trim	-	-	± 5	$\%f_{\text{sirc32k}}$	
$T_{\text{st_sirc32k}}$	Startup time from POR	-	-	20	μs	
$I_{\text{dd_sirc32k}}$	Current consumption	-	2	3	μA	
	Current consumption (Disable)	-	1	-	nA	
T_{Valid}	Valid assertion time ¹	-	50	200	μs	

1. The time is for reference only. It is measured from enable assert to valid bit assert.

5.6 PLL Specification

There are two PLLs on the FC7300F8MDQ series, and their specifications are the same. The PLL0 can be used for the system clock. The PLL reference clock can be selected from FIRC96M (48 MHz output) or FOSC. Refer to the *FC7300F8MDQ Reference Manual* for detailed settings.

Table 23. PLL0 and PLL1 specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
$f_{\text{pll_ref}}$	PLL reference frequency range	2	-	4	MHz	Reference clock after pre-divider
$f_{\text{pll_out}}$	PLL VCO output frequency	240	-	720	MHz	PLL output clock should fall in spec through postdiv
Duty	Duty cycle	45	-	55	%	
I_{pll}	PLL operating current	-	1.5	-	mA	
T_{pll}	Frequency 1% lock time from standby mode	-	60	-	μs	
	Frequency 1% lock time from disabled mode	-	80	-	μs	
T_{lock}	Valid assertion time ¹	-	600	1200	μs	

1. The time is for reference only. It is measured from enable assert to valid bit assert.

Chapter 6 Non-Volatile Memory (NVM)

6.1 NVM Retention

Table 24. NVM retention specification

Symbol	Description	Condition	Min.	Typ.	Max.	Unit	Notes
Program/Erase (P/E) cycles	Number of program/erase cycles per block for 128 KB blocks using Sector Erase	-	100K	-	-	Cycles	
	Number of program/erase cycles per block for 2 MB blocks using Sector Erase	-	1K	-	-	Cycles	
Data Retention	Minimum data retention	Blocks with 0 - 1K P/E cycles	20	-	-	Years	
		Blocks with 100K P/E cycles	10	-	-	Years	

6.2 NVM Program/Erase Time

Table 25. NVM program/erase time¹

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
T _{2wpgm}	2 words (64 bits) program time	-	115	-	μs	
T _{8wpgm}	8 words (256 bits) program time	-	248	-	μs	
T _{32wpgm}	32 words (1024 bits) program time	-	405	-	μs	
T _{2kpgm}	2K bytes sector program time	-	6.45	-	ms	
T _{2kers}	2K bytes sector erase time	-	3.2	23.4 ²	ms	

1. The program/erase time will be influenced by temperature.
2. The max. value is calculated after 20 times of sector erase with retry.

6.3 NVM Max Read Timing

For FC7300F8MDQ, the maximum NVM read frequency is 33.33 MHz.

If the bus clock is configured as 150 MHz, the flash read wait cycle needs to be set to 5 for FC7300F8MDQ.

Chapter 7 Analog

7.1 12-bit SAR ADC

The chip supports up to four 12-bit Successive Approximation (SAR) Analog-to-Digital Converters (ADCs). The specification below is based on minimum chip activity, and the external VRH is used.

7.1.1 ADC Operation Specification

The below is the ADC operation specification.

Table 26. ADC operation specification

Symbol	Description	Min.	Typ. ¹	Max.	Unit	Notes
V _{DDA}	ADC analog supply	3	-	5.5	V	
V _{SSA}	ADC analog ground	-0.1	0	0.1	V	
V _{REFH}	ADC reference high	3	-	5.5	V	<=V _{DDA} 2
V _{REFL}	ADC reference low	-0.1	0	0.1	V	2
V _{ADIN}	ADC input	V _{REFL}	-	V _{REFH}	V	
C _{ADIN}	ADC input capacitance	-	6.4	-	pF	
R _{ADIN}	ADC input resistance	-	1.5	-	KΩ	
R _{AS}	Input source resistance	-	1	-	KΩ	
TUE	Total unadjusted error - Single	-	±4	±8	LSB	3
	Total unadjusted error - Differential	-	±2	±4	LSB	
E _Q	Quantization error	-	±1	-	LSB	
ENOB	Effective number of bits with 1MSPS@12bit mode	-	10.5	-	bits	4
SINAD	Signal-to-noise plus distortion	-	ENOB*6.02 + 1.76	-	dB	@1kHz
F _{ADCK}	ADC conversion clock frequency	6	-	30	MHz	
I _{dd_adc}	Supply current, single mode on V _{REFH}	-	0.62	-	mA	
	Supply current, single mode on V _{DDA}	-	1.7	-	mA	
T _{SU}	Start-up time	-	5	30	μs	5
T _{smp}	Sample time	4/(F _{ADCK})	-	257/(F _{ADCK})		
C _{smp}	Sample cycles	4	-	257	cycles	
C _{conv}	Conversion cycles	-	13	-	cycles	
	Conversion rate	-	1.7	-	MS/s	6

1. Typical values assume V_{REFH} = V_{DDA}, Temp = 25 °C, ADC clock = 30 MHz, source impedance = 100 Ω, and input filter capacitor = 100 nF unless otherwise stated. Typical values are used for reference only, not tested in production.
2. For packages without dedicated V_{REFH} and V_{REFL} pins, V_{REFH} is internally tied to V_{DDA}, and V_{REFL} is internally tied to V_{SSA}.
3. There are dead zones when signals near V_{REFH} and V_{REFL}.
4. If V_{DD} is lower than 4.5 V, the typical value will be 10 bits.

5. The time is for reference only. It is measured from enable assert to valid bit assert.
6. Max. Conversion Rate = Max. ADC Clock Frequency / (Sample Cycles + Conversion Cycles). NOTE: Rounding down to the nearest tenth.
7. Appropriate decoupling capacitors to be used to filter noise on the supplies.
8. VSS and VREFL should be shorted on PCB design.
9. Above spec is for direct channels which is based on 12-bit resolution. Additional mux channels performance may be degraded.
10. The ADC has a dead zone when input is close to VREFL or VREFH. The results here are obtained after calibration and removing the impact of this effect.

7.1.2 ADC Sampling Circuit Specification

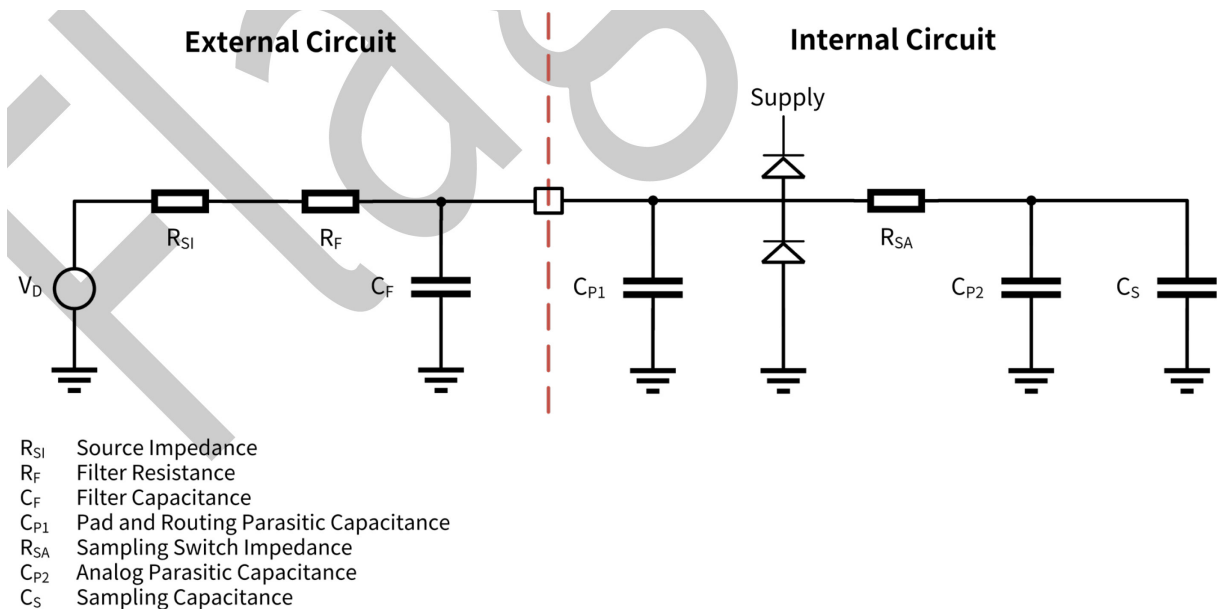
The table below lists the specification for calculating the ADC sampling time.

Table 27. ADC sampling circuit specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
C_S	Sampling capacitance	5.32	6.75	8.2	pF	All channels
R_{SA}	Sampling switch impedance	250	540	1280	Ohm	Channel 0-7
R_{SA}	Sampling switch impedance	490	1040	2320	Ohm	Other channels
C_{P1}	Pin capacitance	1.4	-	5.5	pF	All channels
C_{P2}	Pin capacitance	1.352	-	1.708	pF	All channels

The following figure illustrates the ADC sampling circuit.

Figure 5. ADC sampling circuit



7.2 SDADC Specification

The following table lists the key electrical characteristics and performance parameters of the Sigma-Delta Analog-to-Digital Converter (SDADC).

Table 28. SDADC specification

Description	Min.	Typ.	Max.	Unit	Notes
V _{REFH}	4.5	-	5.5	V	V _{REFH} ≤ V _{DDA}
AVDD	4.5	-	5.5	V	
Analog input range	0	-	AVDD	V	Single-ended
	0	-	2*AVDD		Differential
Input current ¹	-	19.5	-	μA	Gain = 1, Fs = 26.67MHz
On-chip modulator clock frequency	10	26.67	40	MHz	
Signal to Noise Ratio (SNR) for differential input signals	80	-	-	dB	Gain = 1, Fs = 26.67MHz, integrated to Fpb
Spurious-free dynamic range ²	60	-	-	dB	Gain = 1, Fs = 26.67MHz
Gain	1	-	4		
Gain error	-0.2 ³	+/-0.1 ³	-0.2 ³	%	After calibration and V _{REFH} constant after calibration; Fs = 26.67MHz, gain = 1
	-1	-	1	%	Calibrated once; Fs = 25MHz, gain = 1
Power consumption	I _{sdadc}	5	-	mA	One sigma delta ADC, Fs = 26.67MHz
	I _{bias}	1.2	-	mA	One sigma delta ADC bias
DC offset error	-5 ³	-	5 ³	mV	After calibration; Differential mode; Fs = 26.67MHz
	-10	-	10	mV	Calibrated once; Differential mode; Fs = 26.67MHz
Resolution	-	16	-	bits	
Temperature range	-40	-	125	°C	
Common mode voltage bias resistance	-	160	-	Kohm	
Output sampling rate (Fd)	19.531	-	-	KHz	Fd, 19.531K = 10 M/512
Pass band (Fpb)	6.51	-	-	KHz	Fpb is 1/3 of the output sampling rate
Pass band ripple	-	-	-5	dB	CIC3 only, no FIR filter
Stop band attenuation	-	-	-12	dB	0.5...1Fd, CIC3 only, no FIR filter
	-	-	-40	dB	1...1.5Fd, CIC3 only, no FIR filter
	-	-	-40.3	dB	1.5...2Fd, CIC3 only, no FIR filter
	-	-	-53	dB	2...2.5Fd, CIC3 only, no FIR filter
	-	-	-53.5	dB	2.5...OSR/2Fd, CIC3 only, no FIR filter
DC compensation factor	-3	-	-	dB	10 ⁻⁵ Fd, HPF is enabled

1. For detailed information, refer to the *Reference Manual*.
2. This parameter is valid within the range of Fpb.
3. Recalibration needed in case of temperature change ≥ 20 °C.

7.3 CMP Specification

The chip supports high-speed (HS) mode and low-power (LP) mode. The CMP can work in Standby mode.

Table 29. CMP specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
V _{DDA}	CMP analog supply	3	-	5.5	V	
V _{SSA}	CMP analog supply	-0.1	0	0.1	V	
V _{REF_EXT}	DAC external reference high	3	-	5.5	V	
I _{dd_cmp}	CMP operating current with HS mode	-	130	-	μA	
	CMP operating current with LP mode	-	6	-	μA	
V _{input}	Analog input voltage	0	-	V _{DDA}	V	
V _{input_offset}	Analog input offset voltage with HS mode	-	±2	40	mV	
	Analog input offset voltage with LP mode	-	±4	40	mV	
V _{ref_range}	CMP reference voltage	3	-	V _{DDA}	V	
HYS	Analog comparator hysteresis:	-	-	-	-	
	Hyst = 00	-	0	-	mV	
	Hyst = 01	-	15	-	mV	
	Hyst = 10	-	30	-	mV	
	Hyst = 11	-	45	-	mV	
T _{init}	Initialization delay	-	-	30	μs	
T _{prop}	Propagation delay, 100 mV over drive, supply > 3.0 V (HS mode)	-	50	200	ns	
	Propagation delay, 100 mV over drive, supply > 3.0 V (LP mode)	-	1.6	5	μs	
V _{idd_dac}	8-bit DAC current adder	-	10	-	μA	For 5V input
INL	8-bit DAC integral non-linearity	-	±1	±3	LSB	
DNL	8-bit DAC differential non-linearity	-	±0.5	±1	LSB	Guaranteed monotonic

7.4 TMU Specification

The Temperature Monitor Unit (TMU) provides the temperature voltage (TV) output and the temperature flag (TF) output.

The temperature voltage is connected to the ADC0, so the user can measure the voltage to get the current die junction temperature. Refer to *Chapter TMU* in *FC7300F8MDQ Reference Manual* on how to convert the temperature voltage to temperature

Table 30. TMU_TV specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
TMU_TV_I	Current consumption	-	340	-	μA	
TMU_TV_T _{START}	Circuit startup time	-	5	50	μs	
TMU_TV_ACC1	Temperature output error at circuit output (voltage) ^{1, 2, 3}	-5	-	+5	°C	100°C < T _J <= 150°C
TMU_TV_ACC2	Temperature output error at circuit output (voltage) ^{1, 2, 3}	-10	-	+10	°C	-40°C < T _J <= 100°C

Table 30. TMU_TV specification (continued)

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
TMU_TV_T _{ADCSA}	Required ADC sampling time ¹	-	3	-	μs	

1. Required ADC sampling time specified by parameter TMU_TV_T_{ADCSA} needs to be used at the ADC conversion to guarantee the specified accuracy. A smaller sampling time leads to a less accurate result.
2. The temperature sensor measures the junction temperature T_J at the location where it is placed on die. The local T_J is modulated by current and previous active state of the circuit elements on die.
3. The error caused by ADC conversion and provided temperature calculation formula is not included.

The hysteresis range of the temperature flag is as follows:

Table 31. TMU_TF specification

Description		Min.	Typ.	Max.	Unit	Notes
TF hysteresis	TF_125	-	12	-	°C	
	TF_150	-	12	-	°C	
Temperature flag when it over 150°C	TF_150	-	150	-	°C	
Temperature flag when it over 125°C	TF_125	-	125	-	°C	
Initialization time for start up		-	11	-	μs	
Current consumption		-	300	-	μA	

7.5 HRPWM Specification

The following table lists the key electrical characteristics and performance parameters of the High-resolution PWM (HRPWM) module.

Table 32. HRPWM specification

Description	Min.	Typ.	Max.	Unit	Notes
Clock input	-	300	-	MHz	Duty cycle 49 - 51%
Resolution	-	208	-	ps	
Max. phase error		104		ps	
Initialization time	-	10	-	μs	
Current consumption	-	4	-	mA	

7.6 SSI Specification

The following table lists the key electrical characteristics and performance parameters of the Speed and Sensor Interface (SSI) module.

Table 33. SSI specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
I _{dd_cmp}	Quiescent current	-	90	-	μA	
V _{input}	Analog input voltage	0	-	2	V	

Table 33. SSI specification (continued)

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
HYS	Analog comparator hysteresis:	-	-	-	-	
	sel50=1	-	30	-	mV	
	sel50=0	-	15	-	mV	
T _{init}	Initialization delay	-	-	5	μs	
T _{prop}	Propagation delay	-	50	300	ns	

7.7 HSADC Specification

The following table lists the key electrical characteristics and performance parameters of the High-Speed Analog-to-Digital Converter (HSADC).

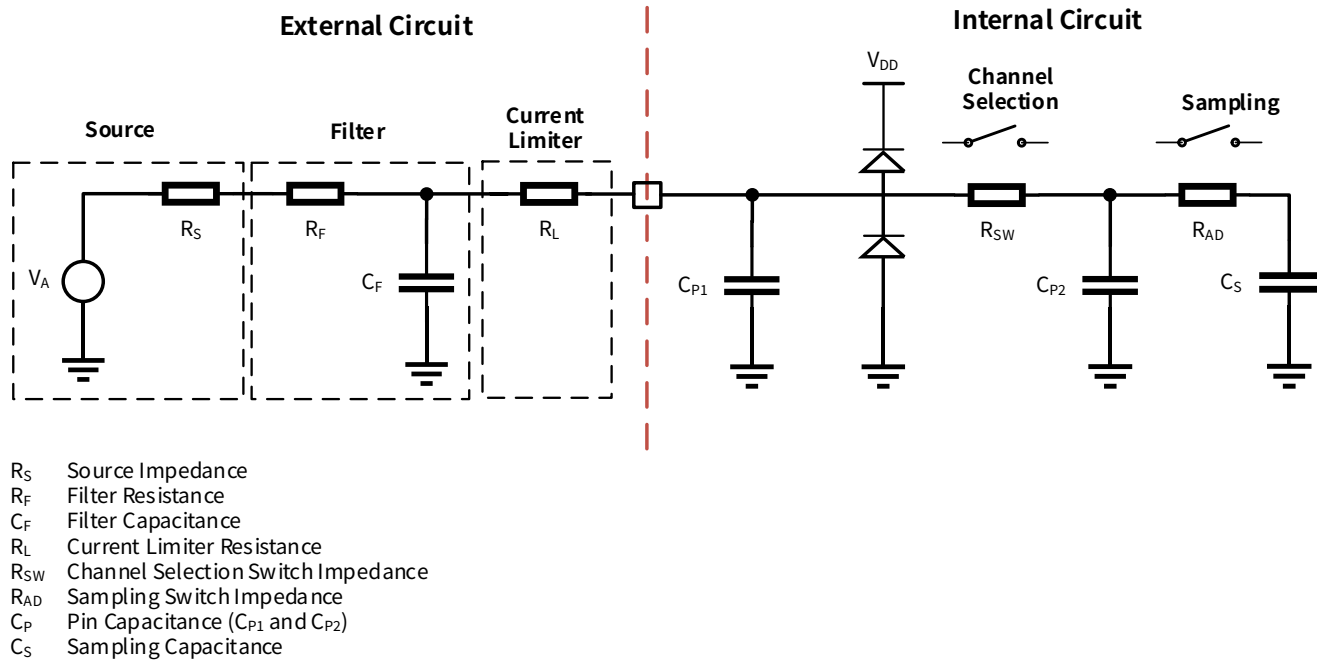
Table 34. HSADC specification

Symbol	Description	Min.	Typ.	Max.	Unit	Notes
V _{DDA}	ADC analog supply	3	-	5.5	V	1, 2
V _{SSA}	ADC analog ground	- 0.1	-	0.1	-	3
V _{REFH}	ADC reference high	3	-	5.5	-	< V _{DDA} 1, 2, 4, 5
V _{REFL}	ADC reference low	- 0.1	-	0.1	-	2, 3, 5
T _{sample}	Sample time	55	-	15000	ns	
f _{CONV}	HSADC conversion operation frequency	6	-	100	MHz	
T _{conv}	Total conversion time	0.75	-	-	μs	
V _{INPUT}	Input voltage range	V _{REFL}	-	V _{REFH}	V	
R _{SW}	Channel selection switch impedance	-	-	350	Ω	Channel [8] and [0]
		-	-	550	Ω	Channel [7:1]
R _{AD}	Sampling switch impedance	-	-	300	Ω	
C _{P1}	Package and PAD capacitance	-	-	2	pF	
C _{P2}	Internal routing and channel switch capacitance	-	-	2	pF	
C _S	Sampling capacitance	-	2.4	3.2	pF	
TUE	Total unadjusted error	- 8	-	8	LSB	6
OS	DC offset error	- 5	-	5	LSB	
E _{GAIN}	Gain Error	- 6	-	6	LSB	
SNR	Signal to Noise Ratio	58	65	-	dB	

1. Typical values assume V_{REFH} = V_{DDA}, Temp = 25 °C. Typical values are used for reference only, not tested in production.
2. Appropriate decoupling capacitors to be used to filter noise on the supplies.
3. V_{SS} and V_{REFL} should be shorted on PCB design.
4. For packages without dedicated V_{REFH} and V_{REFL} pins, V_{REFH} is internally tied to V_{DDA}, and V_{REFL} is internally tied to V_{SSA}.
5. There are dead zones when signals near V_{REFH} and V_{REFL}.
6. The time is for reference only. It is measured from enable assert to valid bit assert.

7. The above specifications are for the direct channels at 12-bit resolution. Performance on additional multiplexed (MUX) channels may be degraded.

Figure 6. HSADC sampling circuit



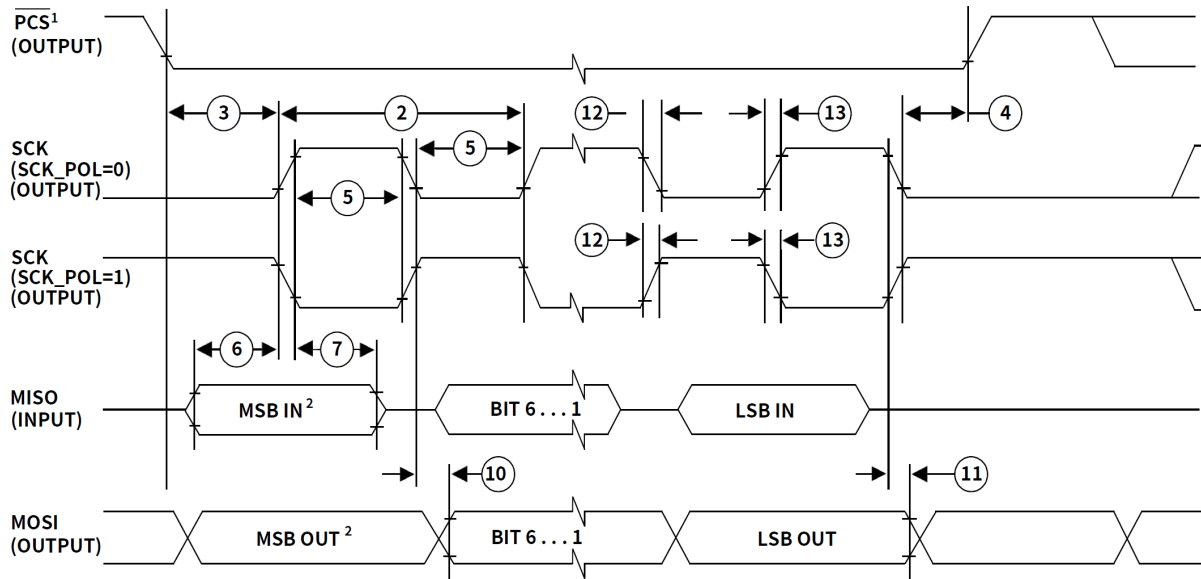
Chapter 8 Peripherals

All peripherals' timings shown below are based on 25 pF (ENET is 15 pF) external loading and input transitions of 1 ns. For normal pad, the SRE is set to 0; for fast pad, the DSE is set to 1. The rise/fall timing is based on 20% VDD_HV to 80% VDD_HV thresholds. Besides, the delay time is based on 50% VDD_HV to 50% VDD_HV thresholds.

8.1 FCSPI Specification

The FCSPI supports a configurable clock by writing the SCK_POL/ SCK_PHA bits, as well as both master and slave mode. The figures below show the timing requirements in different settings.

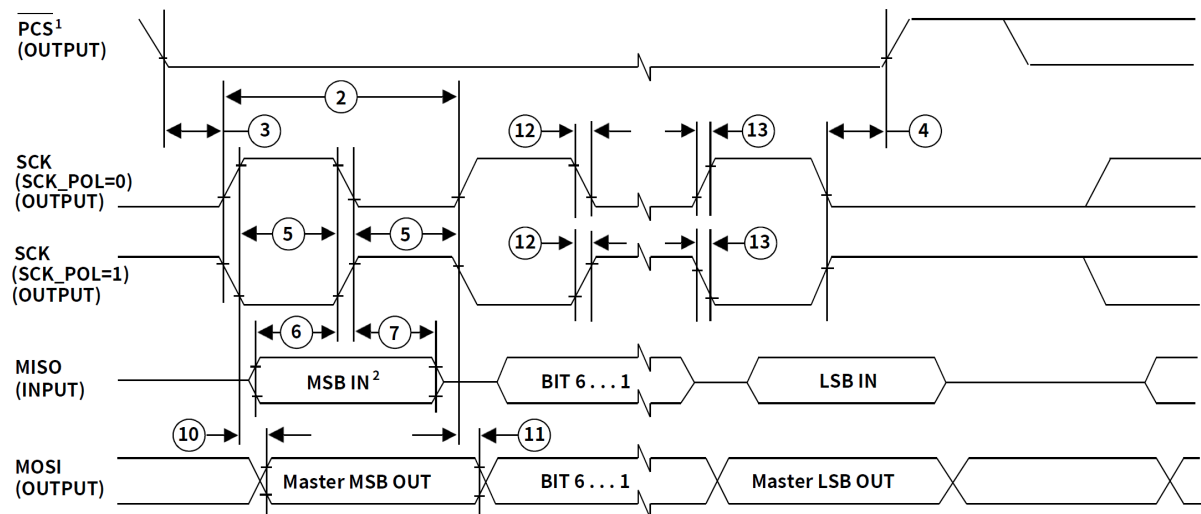
Figure 7. FCSPI master mode timing (CPHA = 0)



Note:

1. If configured as an output.

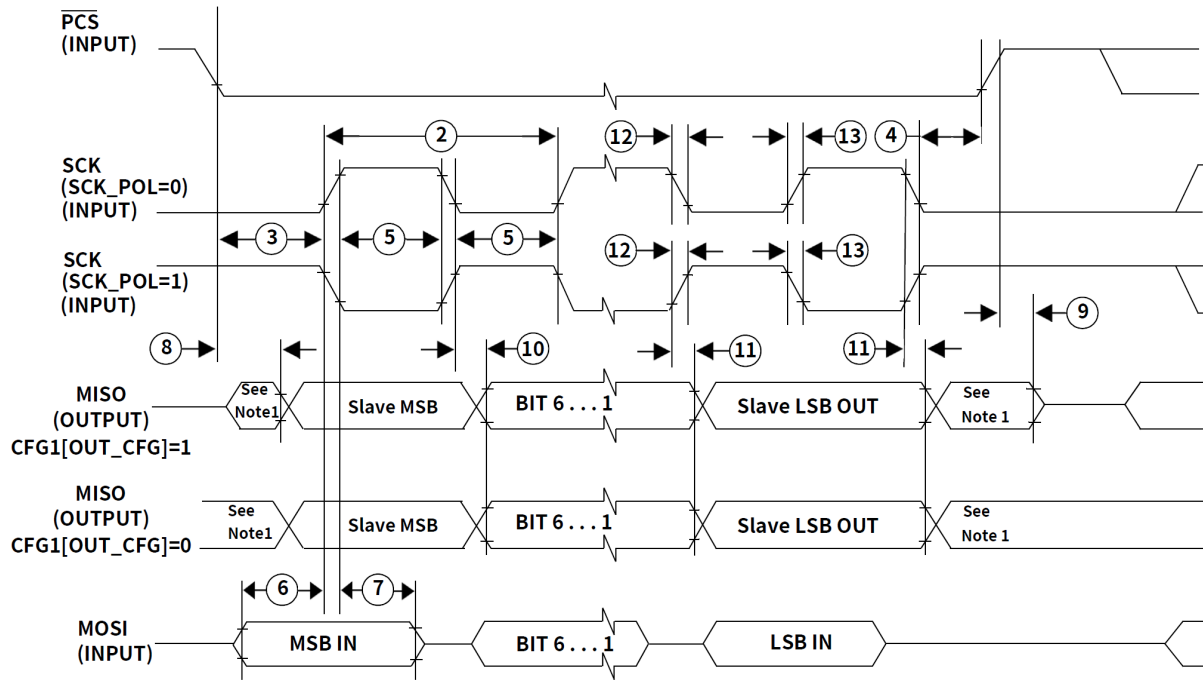
Figure 8. FCSPI master mode timing (CPHA = 1)



Note:

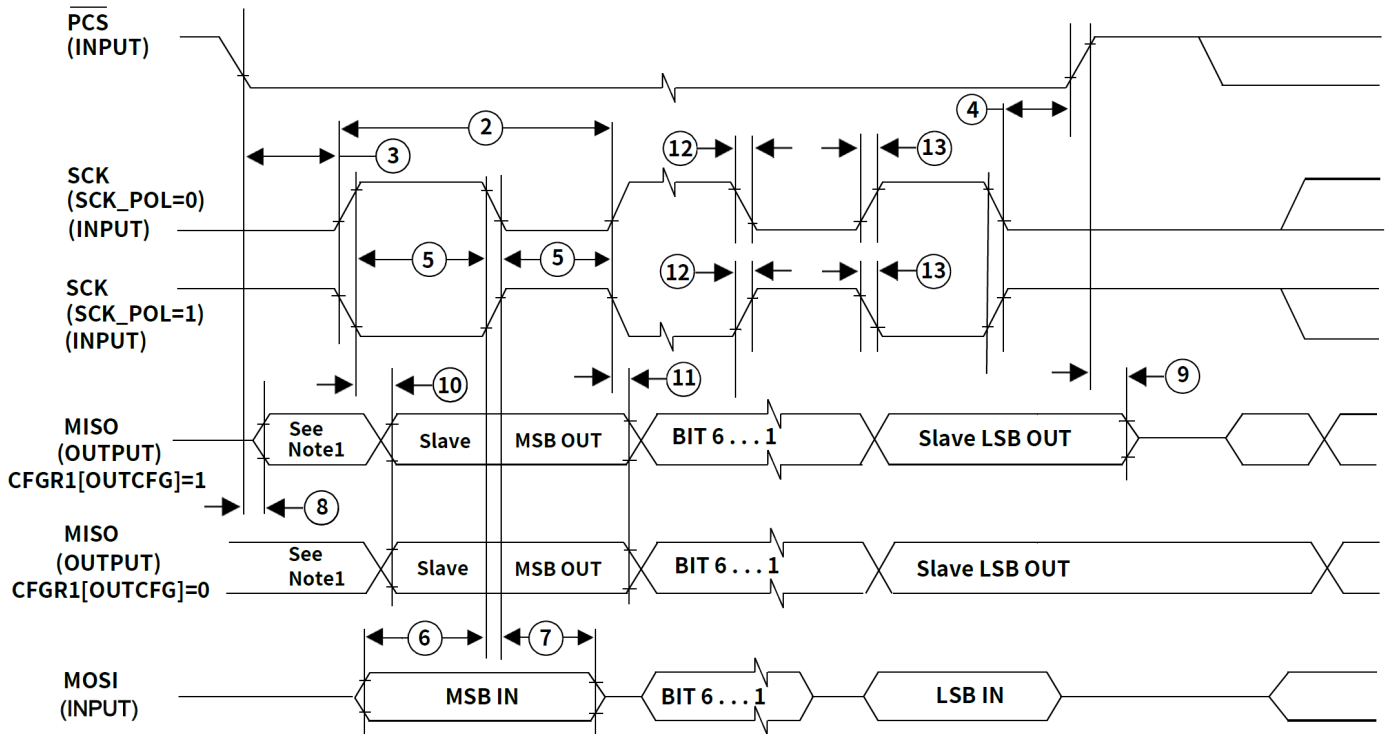
1. If configured as an output.
2. LSBF = 0. For LSBF = 1, bit order is LSB, bit 1, ..., bit 6, MSB.

Figure 9. FCSPI slave mode timing (CPHA = 0)



Note:
1. The bus is driven but may not be equal to the valid serial data being sent.

Figure 10. FCSPI slave mode timing (CPHA = 1)



Note:
1. The bus is driven but may not be equal to the valid serial data being sent.

8.1.1 FCSPI 3V Specification

Below is the FCSPI 3.0-3.6V electrical specification.

Table 35. FCSPI 3V specification

3.0-3.6 V							
Normal Pad (master)							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{op}	Frequency of operation (loopback)	-	-	15	MHz	1
		Frequency of operation (internal clock)	-	-	12.5	MHz	
2	t_{sck}	SCK period (loopback)	66.7	-	-	ns	
		SCK period (internal clock)	80	-	-	ns	
3	t_{Lead}	Enable lead time	-	1/2	-	t_{SCK}	
4	t_{Lag}	Enable lag time	-	1/2	-	t_{SCK}	
5	t_{wsck}	SCK high/low width	-	-	-	ns	
6	t_{SU}	Data Setup time (loopback)	6.2	-	-	ns	
		Data Setup time (internal clock)	30	-	-	ns	
7	t_{HI}	Data Hold time (loopback)	5	-	-	ns	
		Data Hold time (internal clock)	0	-	-	ns	
10	t_v	Data Valid (after SCK edge)	-	-	6	ns	
11	t_{HO}	Data Hold time (outputs)	-10	-	-	ns	
12	t_{RFI}	Rise or Fall time (inputs)	0.5	-	3	ns	
13	t_{RFO}	Rise or Fall time (outputs)	2.8	-	7	ns	
HS Pad (master)							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{op}	Frequency of operation (loopback)	-	-	18.75	MHz	1
		Frequency of operation (internal clock)	-	-	18.75	MHz	
2	t_{sck}	SCK period (loopback)	53.28	-	-	ns	
		SCK period (internal clock)	53.28	-	-	ns	
3	t_{Lead}	Enable lead time	-	1/2	-	t_{SCK}	
4	t_{Lag}	Enable lag time	-	1/2	-	t_{SCK}	
5	t_{wsck}	SCK high/low width	-	-	-	ns	
6	t_{SU}	Data Setup time (loopback)	7.2	-	-	ns	
		Data Setup time (internal clock)	21.5	-	-	ns	
7	t_{HI}	Data Hold time (loopback)	5	-	-	ns	
		Data Hold time (internal clock)	0	-	-	ns	
10	t_v	Data Valid (after SCK edge)	-	-	7	ns	
11	t_{HO}	Data Hold time (outputs)	-10	-	-	ns	
12	t_{RFI}	Rise or Fall time (inputs)	0.5	-	3	ns	
13	t_{RFO}	Rise or Fall time (outputs)	1.8	-	8	ns	

Table 35. FCSPI 3V specification (continued)

Normal Pad (slave)							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{op}	Frequency of operation	-	-	12.5	MHz	1
2	t_{sck}	SCK period	80	-	-	ns	
3	t_{Lead}	Enable lead time	-	1/2	-	t_{sck}	
4	t_{Lag}	Enable lag time	-	1/2	-	t_{sck}	
5	t_{wsck}	SCK high/low width	-	-	-	ns	
6	t_{SU}	Data Setup time	5	-	-	ns	
7	t_{HI}	Data Hold time	4	-	-	ns	
8	t_a	Slave access time	-	-	40	ns	
9	t_{dis}	Slave MISO disable time	-	-	40	ns	
10	t_v	Data Valid (after SCK edge)	-	-	30	ns	
11	t_{HO}	Data Hold time (outputs)	4	-	-	ns	
12	t_{RFi}	Rise or Fall time (inputs)	0.5	-	3	ns	
13	t_{RFo}	Rise or Fall time (outputs)	2.8	-	7	ns	
HS Pad (slave)							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{op}	Frequency of operation	-	-	15	MHz	1
2	t_{sck}	SCK period	66.7	-	-	ns	
3	t_{Lead}	Enable lead time	-	1/2	-	t_{sck}	
4	t_{Lag}	Enable lag time	-	1/2	-	t_{sck}	
5	t_{wsck}	SCK high/low width	-	-	-	ns	
6	t_{SU}	Data Setup time	6	-	-	ns	
7	t_{HI}	Data Hold time	4	-	-	ns	
8	t_a	Slave access time	-	-	33.3	ns	
9	t_{dis}	Slave MISO disable time	-	-	33.3	ns	
10	t_v	Data Valid (after SCK edge)	-	-	23	ns	
11	t_{HO}	Data Hold time (outputs)	4	-	-	ns	
12	t_{RFi}	Rise or Fall time (inputs)	0.5	-	3	ns	
13	t_{RFo}	Rise or Fall time (outputs)	1.8	-	8	ns	

1. The maximum frequencies specified in the table above are measured using the recommended FCSPI pin-mapping options listed in *Section 8.1.3 FCSPI Pin Options*. Performance may vary if alternative pin assignments are used, due to differences in pad characteristics and signal routing

8.1.2 FCSPI 5V Specification

Below is the FCSPI 4.5 to 5.5 V electrical specification.

Table 36. FCSPI 5V specification

4.5-5.5 V							
Normal Pad (master)							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{op}	Frequency of operation (loopback)	-	-	15	MHz	1
		Frequency of operation (internal clock)	-	-	12.5	MHz	
2	t_{sck}	SCK period (loopback)	66.7	-	-	ns	
		SCK period (internal clock)	80	-	-	ns	
3	t_{Lead}	Enable lead time	-	1/2	-	t_{SCK}	
4	t_{Lag}	Enable lag time	-	1/2	-	t_{SCK}	
5	t_{wsck}	SCK high/low width	-	-	-	ns	
6	t_{SU}	Data Setup time (loopback)	7	-	-	ns	
		Data Setup time (internal clock)	30	-	-	ns	
7	t_{HI}	Data Hold time (loopback)	5	-	-	ns	
		Data Hold time (internal clock)	0	-	-	ns	
10	t_v	Data Valid (after SCK edge)	-	-	6	ns	
11	t_{HO}	Data Hold time (outputs)	-10	-	-	ns	
12	t_{RFI}	Rise or Fall time (inputs)	0.5	-	3	ns	
13	t_{RFO}	Rise or Fall time (outputs)	2.8	-	7	ns	
HS Pad (master)							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{op}	Frequency of operation (loopback)	-	-	25	MHz	1
		Frequency of operation (internal clock)	-	-	18.75	MHz	
2	t_{sck}	SCK period (loopback)	40	-	-	ns	
		SCK period (internal clock)	53.28	-	-	ns	
3	t_{Lead}	Enable lead time	-	1/2	-	t_{SCK}	
4	t_{Lag}	Enable lag time	-	1/2	-	t_{SCK}	
5	t_{wsck}	SCK high/low width	-	-	-	ns	
6	t_{SU}	Data Setup time (loopback)	7	-	-	ns	
		Data Setup time (internal clock)	21.5	-	-	ns	
7	t_{HI}	Data Hold time (loopback)	5	-	-	ns	
		Data Hold time (internal clock)	0	-	-	ns	
10	t_v	Data Valid (after SCK edge)	-	-	7	ns	
11	t_{HO}	Data Hold time (outputs)	-10	-	-	ns	
12	t_{RFI}	Rise or Fall time (inputs)	0.5	-	3	ns	
13	t_{RFO}	Rise or Fall time (outputs)	1.8	-	8	ns	

Table 36. FCSPI 5V specification (continued)

Normal Pad (slave)							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{op}	Frequency of operation	-	-	12.5	MHz	1
2	t_{sck}	SCK period	80	-	-	ns	
3	t_{Lead}	Enable lead time	-	1/2	-	t_{sck}	
4	t_{Lag}	Enable lag time	-	1/2	-	t_{sck}	
5	t_{wsck}	SCK high/low width	-	-	-	ns	
6	t_{SU}	Data Setup time	4	-	-	ns	
7	t_{HI}	Data Hold time	4	-	-	ns	
8	t_a	Slave access time	-	-	40	ns	
9	t_{dis}	Slave MISO disable time	-	-	40	ns	
10	t_v	Data Valid (after SCK edge)	-	-	30	ns	
11	t_{HO}	Data Hold time (outputs)	4	-	-	ns	
12	t_{RFi}	Rise or Fall time (inputs)	0.5	-	3	ns	
13	t_{RFo}	Rise or Fall time (outputs)	2.8	-	7	ns	
HS Pad (slave)							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{op}	Frequency of operation	-	-	20	MHz	1
2	t_{sck}	SCK period	50	-	-	ns	
3	t_{Lead}	Enable lead time	-	1/2	-	t_{sck}	
4	t_{Lag}	Enable lag time	-	1/2	-	t_{sck}	
5	t_{wsck}	SCK high/low width	-	-	-	ns	
6	t_{SU}	Data Setup time	6	-	-	ns	
7	t_{HI}	Data Hold time	4	-	-	ns	
8	t_a	Slave access time	-	-	25	ns	
9	t_{dis}	Slave MISO disable time	-	-	25	ns	
10	t_v	Data Valid (after SCK edge)	-	-	20	ns	
11	t_{HO}	Data Hold time (outputs)	4	-	-	ns	
12	t_{RFi}	Rise or Fall time (inputs)	0.5	-	3	ns	
13	t_{RFo}	Rise or Fall time (outputs)	1.8	-	8	ns	

1. The maximum frequencies specified in the table above are measured using the recommended FCSPI pin-mapping options listed in *Section 8.1.3 FCSPI Pin Options*. Performance may vary if alternative pin assignments are used, due to differences in pad characteristics and signal routing

8.1.3 FCSPI Pin Options

The table below lists the recommended pin-mapping options for each FCSPI instance (FCSPI0 to FCSPI7). Each instance supports multiple pin assignment sets, allowing flexible routing of SCK (clock), SIN (input), SOUT (output), and PCSx (chip select) signals to different physical pins. These recommended options ensure that FCSPIs operate at the maximum frequencies specified in tables in sections 8.1.1 and 8.1.2.

Some instances offer both Normal Pad and High-Speed (HS) Pad configurations, ensuring compatibility with the target signal integrity requirements.

Table 37. FCSPI pin options

Instance	Option	SCK	SIN	SOUT	PCS0	PCS1	PCS2	PCS3	Pad Type
FCSPI0	0	PTG23	PTG22	PTF23	PTG21	-	-	-	Normal Pad
	1	PTE0	PTE1	PTA12	PTE25	PTE24	PTE23	PTA11	
FCSPI1	0	PTA28	PTA29	PTE12	PTA27	PTD17	PTA30	PTD16	Normal Pad
	1	PTD2	PTD3	PTD4	PTD23	PTD20	PTA17	PTB17	
FCSPI2	0	PTE13	PTE10	PTA25	PTE11	PTE5	PTI21	PTF24	Normal Pad
	1	PTB31	PTC18	PTC16	PTB23	PTD9	PTD8	PTC15	
	2	PTB3	PTB28	PTB27	PTB25	PTC27	PTF13	PTF14	
FCSPI3	0	PTH21	PTF8	PTG6	PTG7	-	-	-	Normal Pad
	1	PTD6	PTD5	PTD10	PTB22	PTC15	PTC1	PTC0	HS Pad
FCSPI4	0	PTC9	PTA7	PTC30	PTC8	-	-	-	Normal Pad
	1	PTG19	PTG20	PTH18	PTG18	PTF20	PTF19	PTF18	
FCSPI5	0	PTD22	PTF16	PTF17	PTC25	PTB12	PTI12	PTH14	Normal Pad
	1	PTH17	PTI15	PTI16	PTH16	PTI17	-	-	
FCSPI6	0	PTF4	PTF3	PTF5	PTI1	PTI2	PTI0	PTH2	Normal Pad
	1	PTH5	PTH4	PTI3	PTI4	-	-	-	
FCSPI7	0	PTI9	PTH11	PTI10	PTI11	-	-	-	Normal Pad
	1	PTC31	PTA6	PTE7	PTD18	PTD19	PTF15	PTH12	

8.2 MSC Specification

The below shows the MSCx timing diagram and requirements.

Figure 11. MSC timing

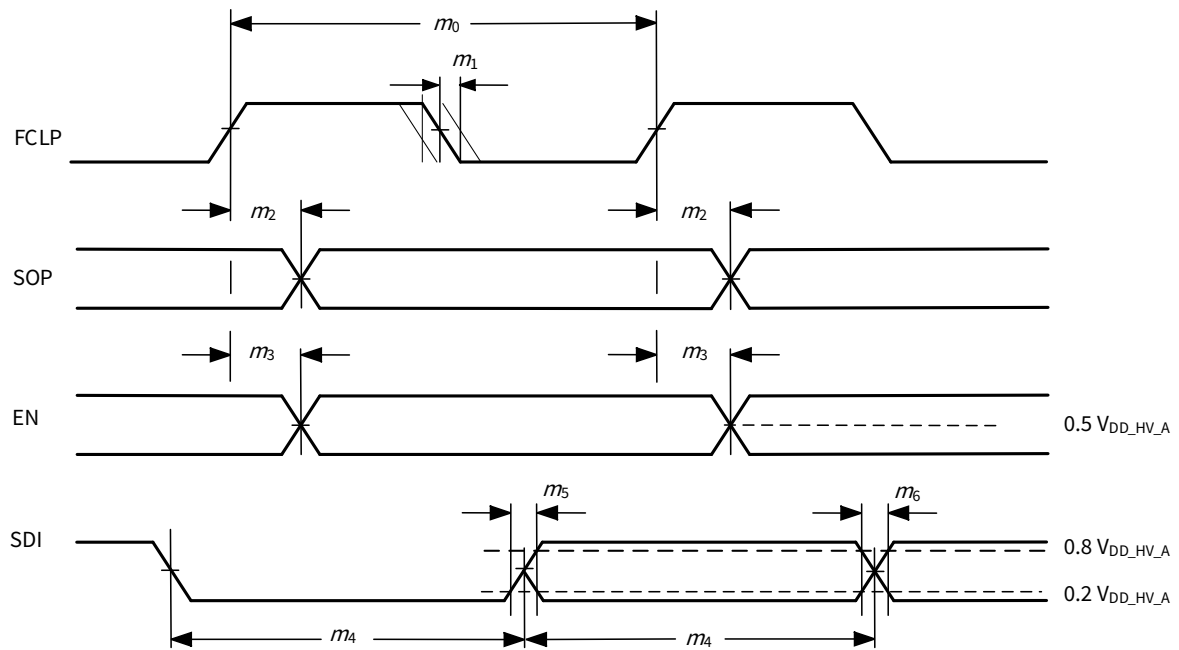


Table 38. MSC electrical specification

MSC Normal Pad							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	F_{MSC}/m_0	Frequency of operation	-	-	25	MHz	
2	m_1	Frequency variation	-4	-	4	ns	
3	m_2, m_3	Output delay	-6	-	6	ns	
4	m_4	Bit time	-	-	$F_{MSC}/8$	MHz	
5	m_5, m_6	Bit rise/fall time	-	-	80	ns	

8.3 ENET Specification

The Ethernet (ENET) supports MII/RMII/RGMII 10/100/1000M specification. For RMII, it supports both 50 MHz and 5MHz RMII clocks. For RGMII, it supports 125 MHz, 25 MHz and 2.5 MHz clock.

Figure 12. MII receive timing diagram

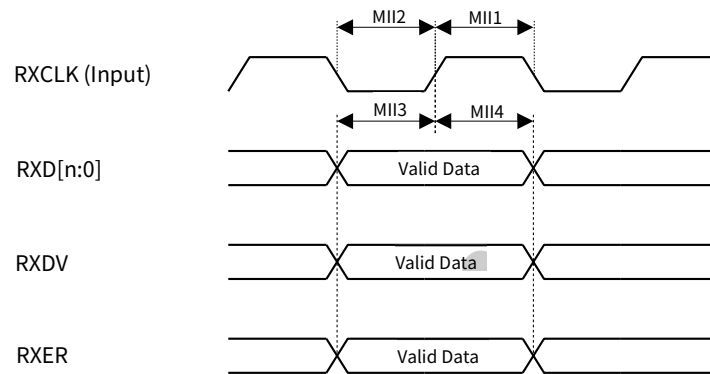


Figure 13. MII transmit timing diagram

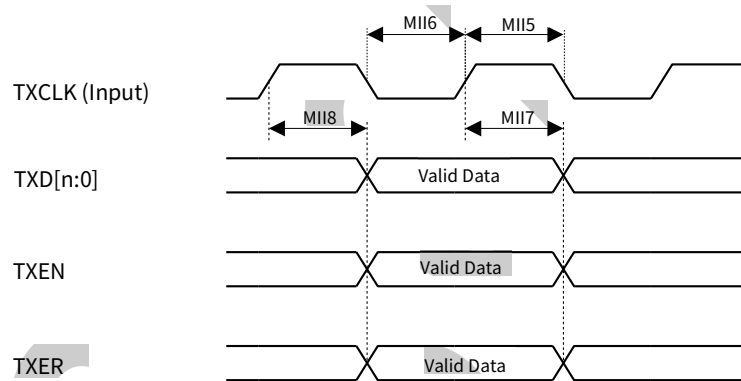


Figure 14. RMII receive timing diagram

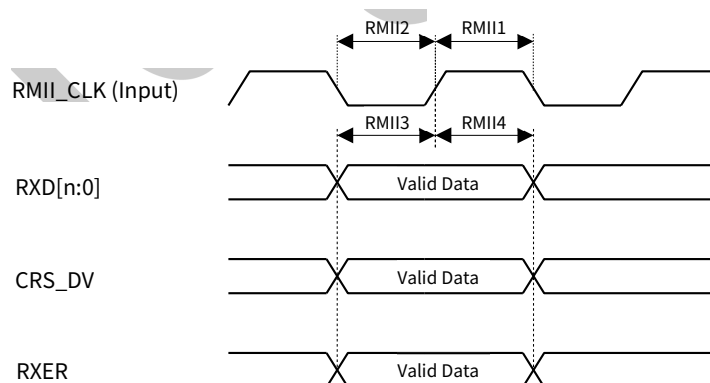


Figure 15. RMII transmit timing diagram

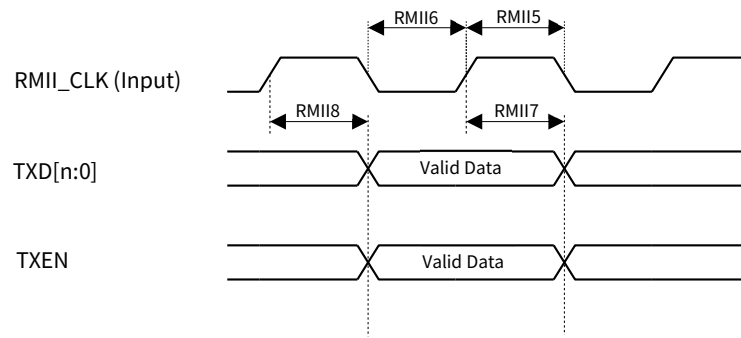


Figure 16. RGMII receive timing diagram

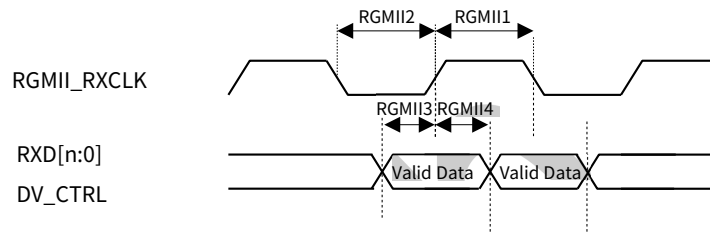


Figure 17. RGMII transmit timing diagram

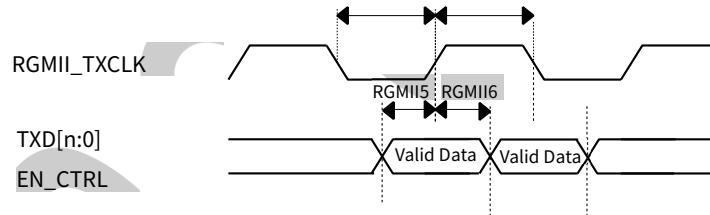


Table 39. MII/RMII electrical specification

MI							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	MII ₁	RXCLK frequency	-	25	-	MHz	
2	MII ₂	RXCLK duty	35%	-	65%	-	
3	MII ₃	RXD[3:0], RXDV, RXER to RXCLK setup	5	-	-	ns	
4	MII ₄	RXD[3:0], RXDV, RXER to RXCLK hold	5	-	-	ns	
5	MII ₅	TXCLK frequency	-	-	25	MHz	
6	MII ₆	TXCLK duty	35%	-	65%	-	
7	MII ₇	TXCLK to TXD[3:0], TXEN, TXER invalid	2	-	-	ns	
8	MII ₈	TXCLK to TXD[3:0], TXEN, TXER valid	-	-	25	ns	
RMII							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	RMII ₁ , RMII ₅	RMII_CLK frequency	-	50	-	MHz	
2	RMII ₂ , RMII ₆	RMII_CLK duty	45%	-	55%	-	
3	RMII ₃	RXD[1:0], CRS_DV, RXER to RMII_CLK setup	4	-	-	ns	
4	RMII ₄	RXD[1:0], CRS_DV, RXER to RMII_CLK hold	2	-	-	ns	
5	RMII ₇	RMII_CLK to TXD[1:0], TXEN invalid	2	-	-	ns	
6	RMII ₈	RMII_CLK to TXD[1:0], TXEN valid	-	-	15	ns	

Table 40. RGMII electrical specification

RGMII							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	RGMII ₁	RGMII_TX/RX CLK frequency	-	125	-	MHz	
2	RGMII ₂	RGMII_RX/TX CLK duty	45%	-	55%	-	
3	RGMII ₃	RXD[3:0], CRS_DV, RXER to RGMII_RX_CLK setup	900	-	2700	ps	
4	RGMII ₄	RGMII_TX_CLK to TXD[31:0], TXEN Jitter	-500	-	500	ns	

The chip also supports the MDC interface for external PHY configuration.

Figure 18. MDIO timing diagram

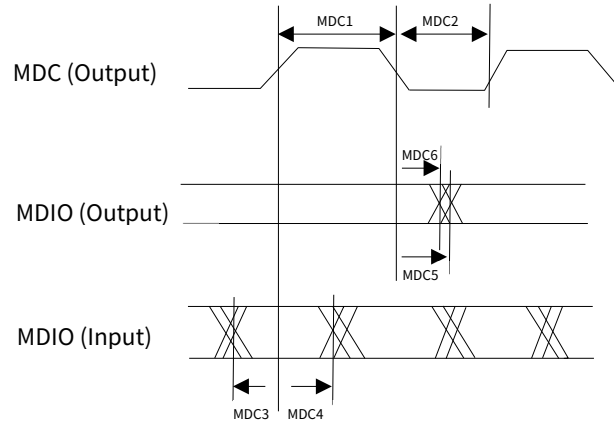


Table 41. MDIO electrical specification

MDIO							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	MDC ₁	MDC clock frequency	-	-	25	MHz	
2	MDC ₂	MDC duty	40%	-	60%	-	
3	MDC ₃	MDIO to MDC rising edge setup	25	-	-	ns	
4	MDC ₄	MDIO to MDC rising edge hold	0	-	-	ns	
5	MDC ₅	MDC falling edge to MDIO output valid	-	-	25	ns	
6	MDC ₆	MDC falling edge to MDIO output invalid	-10	-	-	ns	

8.4 LU Specification

The below shows the timing diagram and requirements for LU emulating SPI communication.

Figure 19. LU timing

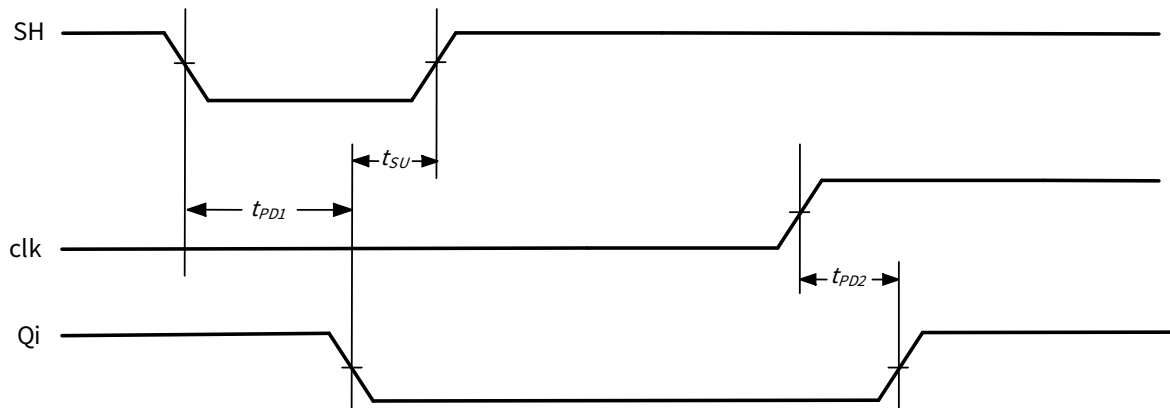


Table 42. LU electrical specification

LU Normal Pad							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{clk}	Frequency of operation (Dual edge)	-	-	15	MHz	
		Frequency of operation (Single edge)	-	-	7.5	MHz	
2	t_{SU}	Data Setup time	5	-	-	ns	
3	t_{PD1}	Propagation delay	-	-	19	ns	
4	t_{PD2}	Propagation delay	-	-	16	ns	

Chapter 9 Debug Modules

The chip supports both JTAG and Serial Wire Debug (SWD) interfaces, and it also has the TPIU interface.

9.1 SWD Specification

Figure 20. SWD clock timing diagram

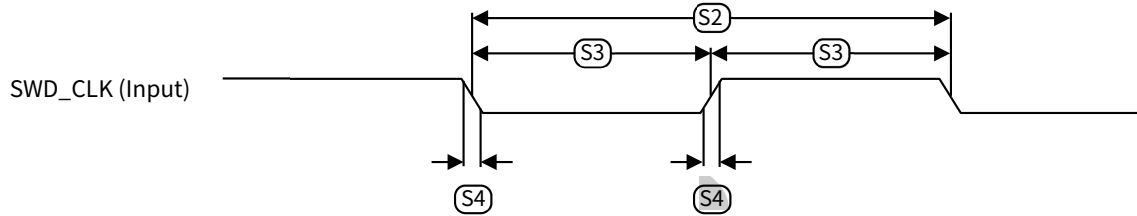


Figure 21. SWD data timing diagram

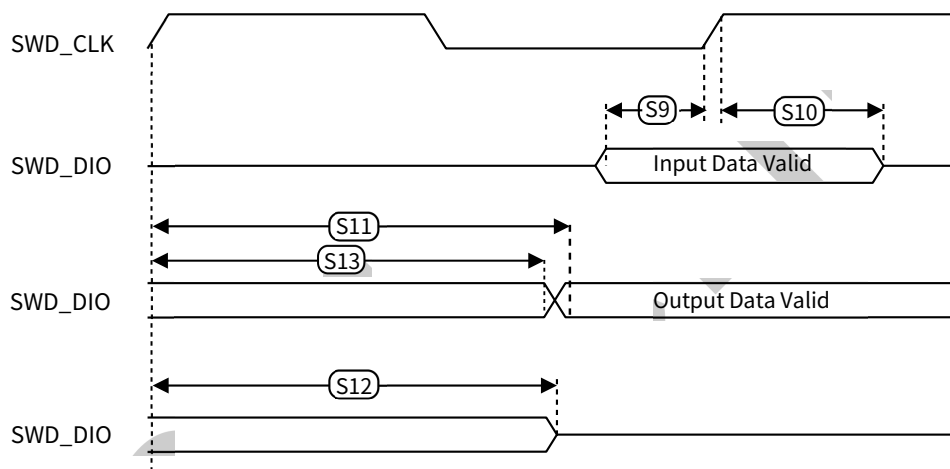


Table 43. SWD electrical specification

SWD							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	S ₁	SWD_CLK frequency	-	-	25	MHz	
2	S ₂	SWD_CLK period	1/S ₁	-	-	ns	
3	S ₃	SWD_CLK clock pulse width	S ₂ /2 - 5	-	S ₂ /2 + 5	ns	
4	S ₄	SWD_CLK rise or fall time	1	-	3	ns	
5	S ₅	SCK high/low width	-	-	-	ns	
6	S ₆	SWD_DIO data setup time	6	-	-	ns	
7	S ₇	SWD_DIO data hold time	3	-	-	ns	
8	S ₈	SWD_CLK to SWD_DIO data valid	-	-	32	ns	
9	S ₉	SWD_CLK to SWD_DIO data high-z	0	-	-	ns	
10	S ₁₀	SWD_CLK to SWD_DIO data invalid	0	-	-	ns	

9.2 Trace Block

Figure 22. Trace block timing diagram

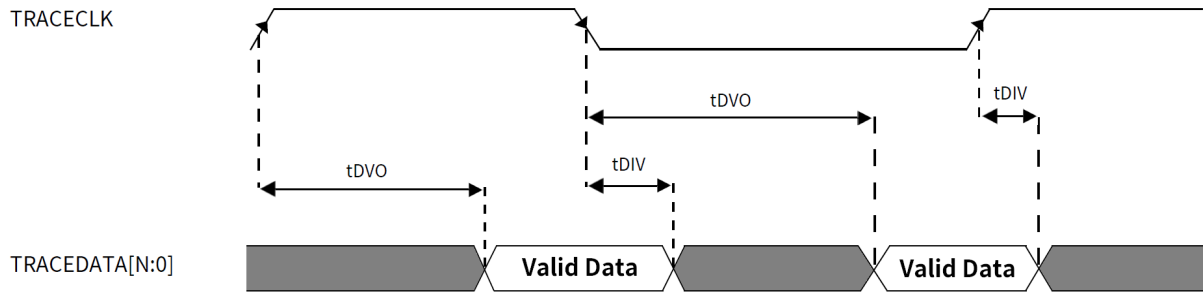


Table 44. Trace block electrical specification

Trace Interface on Normal Pad							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{trace}	Frequency of operation	-	-	25	MHz	
2	t_{dvo}	Data output valid	-	-	9	ns	
3	t_{dvi}	Data output invalid	0	-	-	ns	
Trace Interface on HS Pad							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	f_{trace}	Frequency of operation	-	-	50	MHz	
2	t_{dvo}	Data output valid	-	-	5	ns	
3	t_{dvi}	Data output invalid	0	-	-	ns	

9.3 JTAG Interface

Figure 23. JTAG clock timing diagram

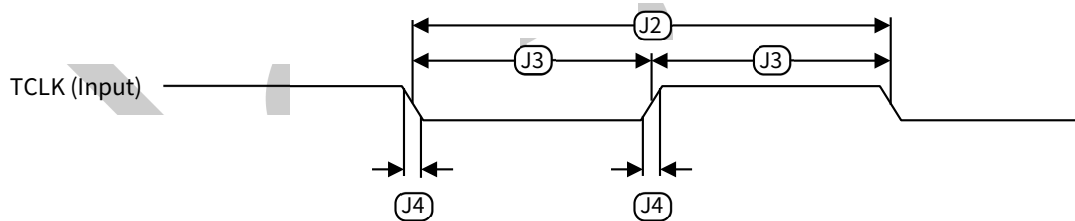


Figure 24. Boundary timing diagram

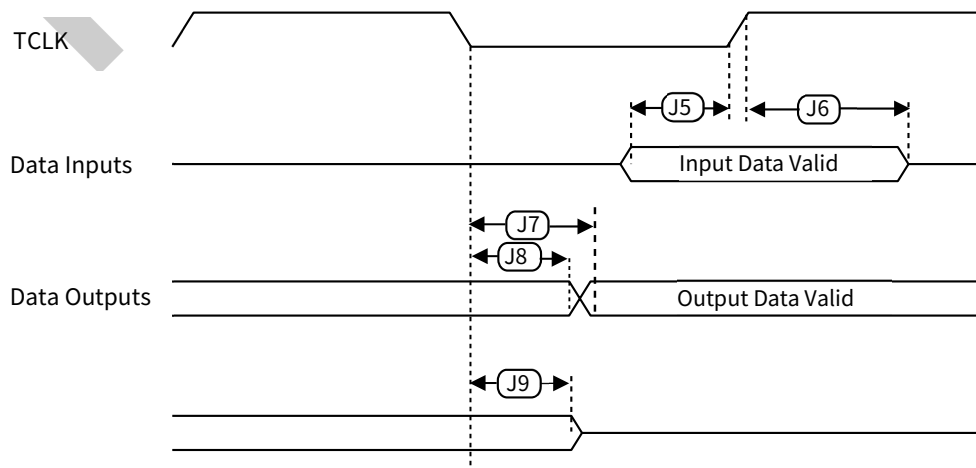
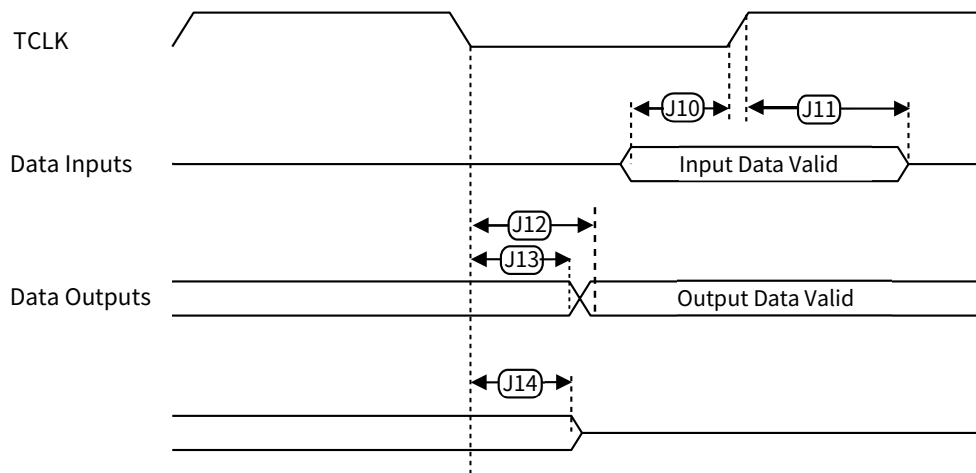


Figure 25. JTAG TAP timing diagram**Table 45. JTAG electrical specification**

JTAG							
Number	Symbol	Description	Min.	Typ.	Max.	Unit	Notes
1	J ₁	TCLK frequency	-	-	15	MHz	
2	J ₂	TCLK period	64	-	-	ns	
3	J ₃	TCLK duty	45%	-	55%	-	
4	J ₄	TCLK rise and fall times	1	-	3	ns	
5	J ₅	Boundary input setup (TCLK)	6	-	-	ns	
6	J ₆	Boundary input hold (TCLK)	2	-	-	ns	
7	J ₇	TCLK low to Boundary output valid	-	-	32	ns	
8	J ₈	TCLK low to Boundary output invalid	0	-	-	ns	
9	J ₉	TCLK low to Boundary output high-Z	-	-	32	ns	
10	J ₁₀	TMS/TDI setup (TCLK)	6	-	-	ns	
11	J ₁₁	TMS/TDI hold (TCLK)	2	-	-	ns	
12	J ₁₂	TCLK low to TDO output valid	-	-	32	ns	
13	J ₁₃	TCLK low to TDO output invalid	0	-	-	ns	
14	J ₁₄	TCLK low to TDO output high-Z			32	ns	

Chapter 10 Package

10.1 Thermal Data

The following table lists the thermal data for the FC7300F8MDQ package. The maximum junction temperature can be calculated based on the package type, operating environment, and PCB design.

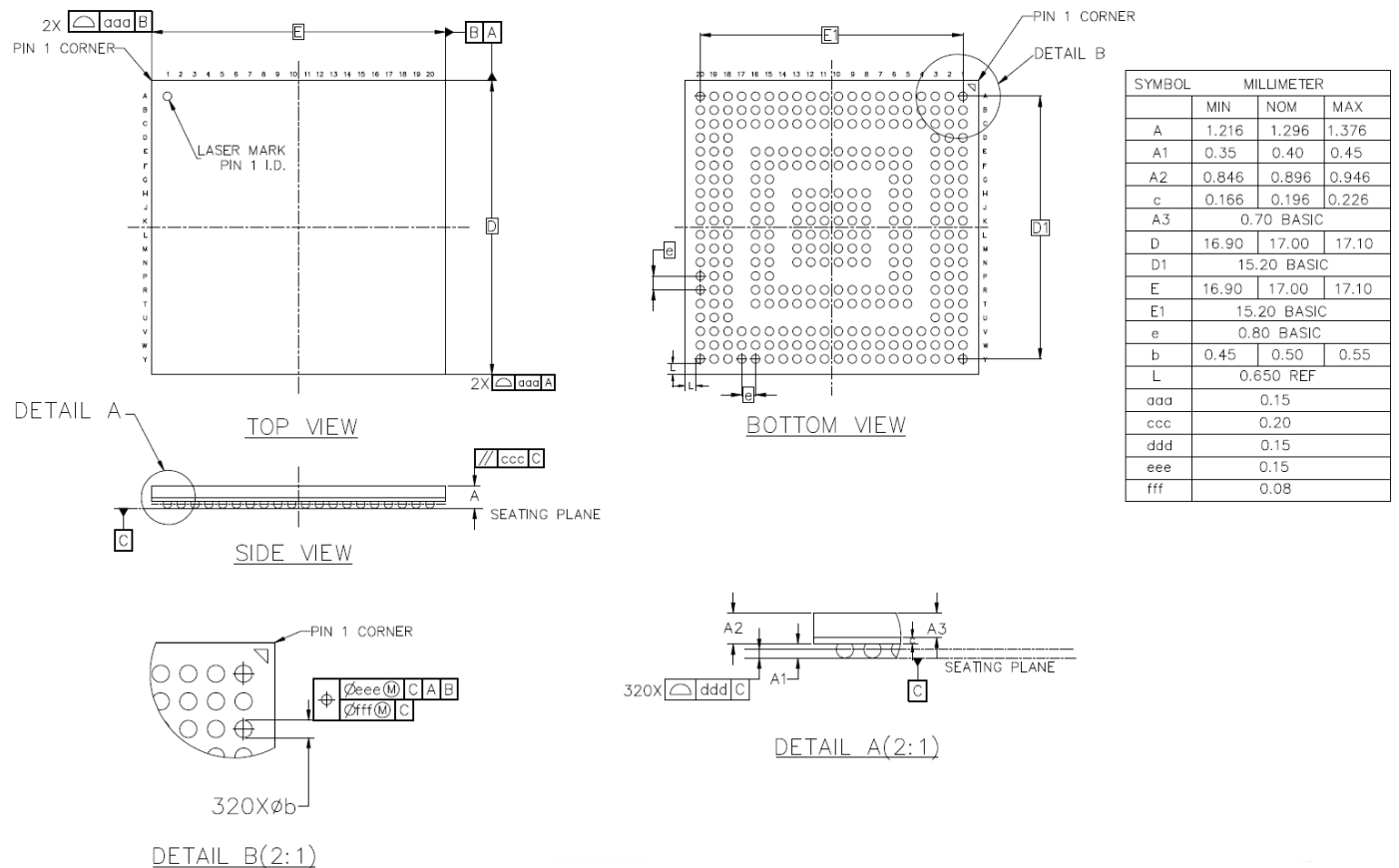
Table 46. Package thermal simulation data

R θ JA	Airflow	PCB Type	Package	Value	Unit
Thermal resistance, Junction to Ambient	Natural Convection	Four-layer board (2s2p)	320LFBGA	20.8	°C/W
	1m/s			18.2	
	2m/s			17.0	

10.2 Package Dimensions

The FC7300F8MDQ is available in 320LFBGA package. The following figure illustrates the package dimensions.

Figure 26. 320LFBGA package outline drawing



Chapter 11 Pinout

11.1 Pinout Diagrams

The following figure illustrates the pinout configurations for the package of the FC7300F8MDQ series.

Figure 27. 320LFBGA pinout (top view)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20							
A	VSS	PTF23	PTA8	PTE26	PTA12	PTE24	PTA13	PTE20	PTI19	PTH19	PTC26	PTH23	PTC6	PTC7	PTD30	PTD29	PTD28	PTD27	PTG17	VSS	A						
B	PTG21	VSS	PTF0	PTA4	RESET_B	PTA11	PTE23	PTE21	PTE2	PTH20	PTE18	PTE17	PTG18	PTF19	PTA0	PTI16	PTB8	PTH15	VSS	PTB9	B						
C	PTE16	PTG22	VSS	PTF1	PTA9	PTC5	PTE1	PTF21	PTE19	PTI20	PTA15	PTH18	PTF20	PTF18	PTI17	PTH16	PTD26	VSS	PTG15	PTG0	C						
D	PTH1	PTA18	PTA19															PTG16	PTB10	PTD2	D						
E	PTE14	PTI0	PTA20	VSS	VDD_HV_A	PTC4	PTE0	PTE25	PTA14	PTI18	PTG20	PTD31	PTI15	PTH17	VSS					PTD25	PTA3	PTC25	E				
F	PTE3	PTI1	PTH2	PTG23	VSS	PTE27	PTA10	PTF22	PTE22	PTE6	PTA16	PTG19	PTA1	VSS	PTF2					PTD24	PTD20	PTD22	F				
G	PTD1	PTA21	PTG2	PTH0	PTG1									PTA2	PTD23					PTD4	PTF17	PTB15	G				
H	PTF4	PTF3	PTI2	PTE15	PTA22	VREFH_0	VDDA_0	V11_3	V11_3	VDD_HV_A	VSS					PTD3	PTD21					PTF16	PTB13	PTH14	H		
J	PTB7	PTE4	PTF5	V11_CFG	V11_0	VREFL_0	VSS	VSS	VSS	VSS	VSS	V11_2					PTB14	PTB12					PTI12	PTH13	PTH12	J	
K	PTB6	PTF24	PTI23	PTB11	V11_0	VSSA_0	VSS	VSS	VSS	VSS	VSS	V11_2					PTB16	PTB17					PTA17	PTF15	PTD19	K	
L	PTI22	PTE10	PTE11	PTA26	V25	VDD_HV_A	VSS	VSS	VSS	VSS	VSS	VDD_HV_A					PTE7	PTD18					PTA6	PTC31	PTC30	L	
M	PTE13	PTA25	PTD0	PTI21	PTE5	VSS	VSS	VSS	VSS	VSS	VSS	VSSA_1					PTC9	PTC29					PTC8	PTC28	PTA7	M	
N	PTA24	PTE12	PTA29	PTA27	PTA23	VSS	VSS	VDD_HV_A	VDDA_1	VREFH_1	VREFL_1					PTG14	PTB3					PTI10	PTI11	PTG13	N		
P	PTA28	PTG3	PTG5	PTF6	PTH4													PTF12	PTF14					PTB27	PTH11	PTI9	P
R	PTG4	PTH3	PTI5	PTI3	VSS	PTA31	PTG7	PTG8	PTH6	V11_1	V11_1	PTC12	PTC21	VSS	VSS					PTB1	PTB25	PTB28	R				
T	PTD17	PTD16	PTA30	VSS	PTE8	VDD_HV_B_0	VSS	VDD_HV_B_1	VSS	VSS	VSS	VDD_HV_A	PTF9	PTG9	PTC22	VSS					PTC27	PTG12	PTB0	T			
U	PTD15	PTE9	PTI4																			PTC24	PTC10	PTF13	U		
V	PTH5	PTB18	VSS	PTB5	PTB30	PTD6	PTD11	PTH7	PTC1	PTD8	PTC17	PTH8	PTB21	PTF10	PTC13	PTG11	PTI14	VSS	PTH10	PTC11	V						
W	PTB19	VSS	PTF8	PTB4	PTC3	PTD5	PTD12	PTB22	PTC0	PTC16	PTB23	PTC14	PTB20	PTI7	PTB29	PTC20	PTG10	PTC23	VSS	PTH9	W						
Y	VSS	PTH21	PTG6	PTB26	PTC2	PTD7	PTD10	PTC15	PTD9	PTB31	PTC18	PTH22	PTB24	PTI6	PTI8	PTC19	PTF11	PTI13	PTF7	VSS	Y						
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20							

11.2 Pin Definitions

The following table describes the default function for each pin. For a complete list of alternate functions and their configurations, refer to the pinout attachment in the *FC7300F8MDQ Reference Manual*.

NOTE:

- 1) The IOs highlighted in green belong to the VDD_HV_B domain. Unlike the rest IOs, they can operate with independent power supplies.
- 2) The IOs highlighted in orange are connected to the exposed pads.

Table 47. FC7300F8MDQ pinout definition

320LFBGA	Pin Name	Default Function
C2	PTG22	ADC2_SE5
F5	PTG23	ADC2_SE1
H9	VDDA_0	VDDA_0
H8	VREFH_0	VREFH_0
J8	VREFL_0	VREFL_0
K8	VSSA_0	VSSA_0
D2	PTA18	HSADC0_CH0
D3	PTA19	HSADC0_CH1
E3	PTA20	HSADC1_CH0
C1	PTE16	HSADC1_CH1
G5	PTH0	HSADC0_CH2
D1	PTH1	HSADC0_CH3
F3	PTH2	ADC2_SE4
E2	PTI0	ADC2_SE0
E1	PTE14	OSC32K_XTAL
F1	PTE3	OSC32K_EXTAL
G6	PTG1	ADC0_SE0
G3	PTG2	ADC0_SE4
H6	PTA22	ADC1_SE0
F2	PTI1	HSADC1_CH2
H5	PTE15	HSADC1_CH3
G2	PTA21	ADC1_SE4
G1	PTD1	HSADC0_CH4
H3	PTI2	HSADC0_CH5
H2	PTF3	ADC0_SE1
H1	PTF4	HSADC1_CH4
J3	PTF5	HSADC1_CH5
J2	PTE4	ADC0_SE5
J5	V11_CFG	V11_CFG
K5	PTB11	DISABLED
J6	V11_0	V11
K6	V11_0	V11
VSS	VSS	VSS
VDD_HV_A	VDD_HV_A	VDD_HV_A
L6	V25	V25
J1	PTB7	EXTAL
K1	PTB6	XTAL
L5	PTA26	ADC0_SE2
K3	PTI23	ADC0_SE6
K2	PTF24	ADC1_SE1

Table 46. FC7300F8MDQ pinout definition (continued)

320LFBGA	Pin Name	Default Function
M5	PTI21	ADC1_SE5
L1	PTI22	ADC1_SE2
L3	PTE11	ADC1_SE6
L2	PTE10	ADC0_SE3
M1	PTE13	ADC0_SE7
M2	PTA25	ADC1_SE3
M6	PTE5	ADC1_SE7
M3	PTD0	ADC0_SE8
N6	PTA23	ADC1_SE8/CMP0_IN0
N1	PTA24	ADC0_SE9/CMP0_IN1
N5	PTA27	ADC1_SE9
N2	PTE12	ADC0_SE10
P1	PTA28	ADC1_SE10
N3	PTA29	ADC0_SE11
P2	PTG3	ADC1_SE11
R1	PTG4	ADC0_SE12/CMP0_IN2
P3	PTG5	ADC1_SE12
R2	PTH3	ADC0_SE13
VSS	VSS	VSS
VDD_HV_A	VDD_HV_A	VDD_HV_A
R3	PTI5	ADC1_SE13
P5	PTF6	ADC0_SE14
T1	PTD17	ADC1_SE14/CMP0_IN3
T3	PTA30	ADC0_SE15/CMP0_IN4
T2	PTD16	ADC1_SE15/CMP0_IN5
U1	PTD15	ADC0_SE16/CMP0_IN6
U2	PTE9	ADC1_SE16/CMP0_IN7
P6	PTH4	SSI0_ANA_CH0
V1	PTH5	SSI0_ANA_CH1
R5	PTI3	SSI0_ANA_CH2
U3	PTI4	SSI0_ANA_CH3
R7	PTA31	ADC0_SE17/ADC1_SE17
V2	PTB18	ADC0_SE18/ADC1_SE18
W1	PTB19	ADC0_SE19/ADC1_SE19
VSS	VSS	VSS
Y2	PTH21	DISABLED
W3	PTF8	DISABLED
Y3	PTG6	DISABLED
R8	PTG7	DISABLED
T6	PTE8	DISABLED

Table 46. FC7300F8MDQ pinout definition (continued)

320LFBGA	Pin Name	Default Function
V4	PTB5	DISABLED
W4	PTB4	DISABLED
Y4	PTB26	DISABLED
V5	PTB30	DISABLED
W5	PTC3	DISABLED
Y5	PTC2	DISABLED
Y6	PTD7	DISABLED
VSS	VSS	VSS
T7	VDD_HV_B_0	VDD_HV_B
V6	PTD6	DISABLED
W6	PTD5	DISABLED
W7	PTD12	DISABLED
V7	PTD11	DISABLED
Y7	PTD10	DISABLED
R9	PTG8	DISABLED
R10	PTH6	DISABLED
V8	PTH7	DISABLED
W8	PTB22	DISABLED
Y8	PTC15	DISABLED
VSS	VSS	VSS
T9	VDD_HV_B_1	VDD_HV_B
V9	PTC1	DISABLED
W9	PTC0	DISABLED
Y9	PTD9	DISABLED
V10	PTD8	DISABLED
V11	PTC17	DISABLED
W10	PTC16	DISABLED
Y10	PTB31	DISABLED
Y11	PTC18	DISABLED
W11	PTB23	DISABLED
W12	PTC14	DISABLED
V12	PTH8	DISABLED
Y12	PTH22	DISABLED
VDD_HV_A	VDD_HV_A	VDD_HV_A
R11	V11_1	V11
R12	V11_1	V11
VSS	VSS	VSS
W13	PTB20	ADC5_SE0
V13	PTB21	ADC5_SE4
Y13	PTB24	ADC5_SE1

Table 46. FC7300F8MDQ pinout definition (continued)

320LFBGA	Pin Name	Default Function
Y14	PTI6	ADC5_SE5
W14	PTI7	ADC4_SE0
Y15	PTI8	ADC4_SE4
T13	PTF9	ADC4_SE1
V14	PTF10	ADC4_SE5
W15	PTB29	ADC3_SE0
V15	PTC13	SDADC0_CH0_P0
R13	PTC12	SDADC0_CH0_N0
Y16	PTC19	ADC3_SE4
W16	PTC20	SDADC0_CH1_P0
R14	PTC21	SDADC0_CH1_N0
Y17	PTF11	SDADC_CH_P1/ADC5_SE2
T14	PTG9	SDADC_CH_N1/ADC4_SE2
W17	PTG10	SDADC0_CH2_P0
V16	PTG11	SDADC0_CH2_N0
VSS	VSS	VSS
VDD_HV_A	VDD_HV_A	VDD_HV_A
N11	VDDA_1	VDDA_1
N12	VREFH_1	VREFH_1
N13	VREFL_1	VREFL_1
M13	VSSA_1	VSSA_1
T15	PTC22	SDADC_CH_P2/ADC5_SE6
Y18	PTI13	SDADC1_CH0_P0
V17	PTI14	ADC3_SE1
W18	PTC23	SDADC_CH_N2/ADC4_SE6
Y19	PTF7	SDADC1_CH0_N0
W20	PTH9	ADC3_SE5
V19	PTH10	SDADC1_CH1_P0
U18	PTC24	SDADC1_CH1_N0
V20	PTC11	SDADC_CH_P3/ADC3_SE2
U19	PTC10	SDADC_CH_N3/ADC3_SE6
T18	PTC27	ADC5_SE3
U20	PTF13	SDADC1_CH2_P0
P15	PTF12	ADC5_SE7
P16	PTF14	SDADC1_CH2_N0
T19	PTG12	ADC4_SE3
VSS	VSS	VSS
VDD_HV_A	VDD_HV_A	VDD_HV_A
R18	PTB1	ADC4_SE7
T20	PTB0	ADC3_SE3

Table 46. FC7300F8MDQ pinout definition (continued)

320LFBGA	Pin Name	Default Function
R19	PTB25	ADC3_SE7
N16	PTB3	ADC5_SE8
P18	PTB27	ADC4_SE8
R20	PTB28	ADC3_SE8
P19	PTH11	ADC5_SE9
P20	PTI9	ADC4_SE9
N18	PTI10	ADC3_SE9
N19	PTI11	ADC5_SE10
N20	PTG13	ADC4_SE10
N15	PTG14	ADC3_SE10
M19	PTC28	ADC5_SE11
M15	PTC9	ADC4_SE11
M18	PTC8	ADC3_SE11
M16	PTC29	ADC5_SE12
M20	PTA7	ADC4_SE12
L20	PTC30	ADC3_SE12
L18	PTA6	ADC5_SE13
L19	PTC31	ADC4_SE13
L15	PTE7	ADC3_SE13
L16	PTD18	ADC5_SE14
K20	PTD19	ADC4_SE14
VSS	VSS	VSS
VDD_HV_A	VDD_HV_A	VDD_HV_A
K13	V11_2	V11
J13	V11_2	V11
K19	PTF15	ADC3_SE14
K18	PTA17	ADC5_SE15
K16	PTB17	ADC4_SE15
J20	PTH12	ADC3_SE15
J19	PTH13	ADC5_SE16
H20	PTH14	ADC4_SE16
J18	PTI12	ADC3_SE16
K15	PTB16	ADC5_SE17
G20	PTB15	ADC4_SE17
J15	PTB14	ADC3_SE17
H19	PTB13	ADC5_SE18
J16	PTB12	ADC4_SE18
F20	PTD22	ADC3_SE18
H18	PTF16	ADC5_SE19
G19	PTF17	ADC4_SE19

Table 46. FC7300F8MDQ pinout definition (continued)

320LFBGA	Pin Name	Default Function
E20	PTC25	ADC3_SE19
F19	PTD20	ADC5_SE20
H16	PTD21	ISP_EN_B
G18	PTD4	ADC4_SE20
H15	PTD3	ADC3_SE20
D20	PTD2	ADC5_SE21
G16	PTD23	ADC4_SE21
VSS	VSS	VSS
VDD_HV_A	VDD_HV_A	VDD_HV_A
E19	PTA3	ADC3_SE21
G15	PTA2	ADC5_SE22
F18	PTD24	ADC4_SE22
F16	PTF2	ADC3_SE22
C20	PTG0	ADC5_SE23
D19	PTB10	ADC4_SE23
E18	PTD25	ADC3_SE23
B20	PTB9	DISABLED
C19	PTG15	DISABLED
D18	PTG16	DISABLED
A19	PTG17	DISABLED
B18	PTH15	DISABLED
C17	PTD26	DISABLED
A18	PTD27	DISABLED
B17	PTB8	DISABLED
A17	PTD28	DISABLED
C16	PTH16	DISABLED
E15	PTH17	DISABLED
E14	PTI15	DISABLED
B16	PTI16	DISABLED
C15	PTI17	DISABLED
F14	PTA1	DISABLED
A16	PTD29	DISABLED
B15	PTA0	DISABLED
A15	PTD30	DISABLED
VSS	VSS	VSS
VDD_HV_A	VDD_HV_A	VDD_HV_A
C14	PTF18	DISABLED
B14	PTF19	DISABLED
C13	PTF20	DISABLED
E13	PTD31	DISABLED

Table 46. FC7300F8MDQ pinout definition (continued)

320LFBGA	Pin Name	Default Function
A14	PTC7	DISABLED
A13	PTC6	DISABLED
B13	PTG18	DISABLED
F13	PTG19	ADC2_SE23
E12	PTG20	ADC2_SE22
C12	PTH18	ADC2_SE21
B12	PTE17	DISABLED
F12	PTA16	DISABLED
A12	PTH23	ADC2_SE20
A11	PTC26	ADC2_SE19
B11	PTE18	ADC2_SE18
C11	PTA15	ADC2_SE17
A10	PTH19	ADC2_SE16
B10	PTH20	ADC2_SE15
E11	PTI18	ADC2_SE14
A9	PTI19	ADC2_SE13
F11	PTE6	ADC2_SE12
B9	PTE2	ADC2_SE11
C10	PTI20	ADC2_SE10
VSS	VSS	VSS
VDD_HV_A	VDD_HV_A	VDD_HV_A
H11	V11_3	V11
H10	V11_3	V11
C9	PTE19	ADC2_SE9
A8	PTE20	ADC2_SE8
E10	PTA14	DISABLED
B8	PTE21	DISABLED
F10	PTE22	DISABLED
A7	PTA13	DISABLED
C8	PTF21	HSADC3_CH5
F9	PTF22	HSADC3_CH4
B7	PTE23	ADC2_SE7
A6	PTE24	ADC2_SE3
E9	PTE25	HSADC2_CH5
A5	PTA12	HSADC2_CH4
B6	PTA11	NMI_b
F8	PTA10	JTAG_TDO
C7	PTE1	HSADC3_CH3
E8	PTE0	HSADC3_CH2
A4	PTE26	HSADC2_CH1

Table 46. FC7300F8MDQ pinout definition (continued)

320LFBGA	Pin Name	Default Function
B5	RESET_b	RESET_b
C6	PTC5	JTAG_TDI
E7	PTC4	JTAG_TCLK/SWD_CLK
B4	PTA4	JTAG_TMS/SWD_DIO
F7	PTE27	HSADC2_CH0
C5	PTA9	HSADC3_CH1
A3	PTA8	HSADC3_CH0
B3	PTF0	HSADC2_CH3
C4	PTF1	HSADC2_CH2
VSS	VSS	VSS
VDD_HV_A	VDD_HV_A	VDD_HV_A
A2	PTF23	ADC2_SE6
B1	PTG21	ADC2_SE2

Revision History

Revision	Date	Changes
0.1	2024/05/27	Preliminary version
0.1.1	2024/08/20	- Removed the typical value of "f_{osc}" from <i>Table 19. FOSC specification of Section 5.1 FOSC Specification</i>; - Updated the HYS-related data in <i>Table 34. SSI electrical specification of Section 7.7 SSI Specification</i>; - In <i>Section 8.1 FCSPi Specification</i>: - In <i>Table 35. FCSPi 3V specification</i>: - Changed the max. value of "t_{su} (Data Setup time (loopback))" on Normal Pad (master) from "5.5" to "6.2"; - Changed the max. value of "t_{RFI}" on Normal Pad (master) from "3" to "15"; - Changed the max. value of "t_{su} (Data Setup time (loopback))" on HS pad (master) from "5" to "7.2"; - Changed the max. value of "t_{su} (Data Setup time (internal clock))" on HS pad (master) from "20.5" to "21.5"; - Changed the max. value of "t_v" on HS pad (master) from "4.5" to "7"; - Changed the max. value of "t_{su}" on Normal Pad (slave) from "4.5" to "5"; - Changed the max. value of "t_{su}" on HS pad (slave) from "4" to "6"; - Changed the max. value of "t_v" on HS pad (slave) from "20" to "23". - In <i>Table 36. FCSPi 5V specification</i>: - Changed the max. value of "t_{su} (Data Setup time (loopback))" on Normal pad (master) from "5" to "7"; - Changed the max. value of "t_v" on Normal pad (master) from "5" to "6"; - Changed the max. value of "t_{su} (Data Setup time (loopback))" on HS pad (master) from "4" to "7"; - Changed the max. value of "t_{su} (Data Setup time (internal clock))" on HS pad (master) from "20" to "21.5"; - Changed the max. value of "t_v" on HS pad (master) from "4" to "7"; - Changed the max. value of "t_{su}" on HS pad (slave) from "4" to "6". - In <i>Table 42. Trace block electrical specification of Section 9.2 Trace Block</i>: - Changed the max. value of "t_{dvo}" on Normal Pad from "5" to "9"; - Changed the max. value of "f_{trace}" on HS Pad from "75" to "50"; - Changed the max. value of "t_{dvo}" on HS Pad from "3" to "6"; - Changed the min. value of "t_{dvi}" on HS Pad from "-1" to "0".
0.1.2	2024/10/30	- Updated the max. functional frequencies of modules in <i>Table 4. Max. functional clock of modules in Section 3.4 Clock Operating Condition</i>; - Added pinout diagrams and table to <i>Chapter 11 Pinout</i>
0.2	2025/01/20	- Removed HSADC-related specification; - Added notes on VDDA and VDD_HV_A under <i>Table 3. Operating requirements to Section 3.2 Operation Condition</i>

Revision	Date	Changes
0.3	2025/05/14	- Updated the max. functional frequencies of modules in <i>Table 4. Max. functional clock of modules in Section 3.4 Clock Operating Condition</i>; - In <i>Chapter 5 Clock Specification</i>: - Added the "T_{valid}" related data and the corresponding note to <i>Table 20. FIRC96M specification in Section 5.3 FIRC96M Specification</i>; - Added the "T_{valid}" related data and the corresponding note to <i>Table 21. SIRC12M specification in Section 5.4 SIRC12M Specification</i>; - Added the "T_{valid}" related data and the corresponding note to <i>Table 22. SIRC32k specification in Section 5.5 SIRC32k Specification</i>; - Added the "T_{lock}" related data and the corresponding note to <i>Table 23. PLL0 and PLL1 specification in Section 5.6 PLL Specification</i>. - In <i>Chapter 7 Analog</i>: - Added the max. value of "T_{su}" to <i>Table 25. ADC operation specification in Section 7.1.1 ADC Operation Specification</i>; - Updated the data in <i>Table 27. SDADC Specification in Section 7.2 SDADC Specification</i>; - Added the max. value of "TMU_TV_T_{START}" to <i>Table 29. TMU_TV specification in Section 7.4 TMU Specification</i>. - In <i>Section 8.1 FCSPI Specification</i>: - Removed the max. data and added the min. data of "T_{su}" in Normal Pad (master), and HS Pad (master) in both <i>Table 33. FCSPI 3V specification</i> and <i>Table 34. FCSPI 5V specification</i>; - Changed the max. value of "t_{RFI}" from "15" to "3" in Normal Pad (master) in <i>Table 33. FCSPI 3V specification</i>. - Added a missed pin "VSSA_1" to <i>Table 43. FC7300F8MDQ pinout in Chapter 11 Pinout</i>; - Fixed typos
0.4	2026/01/08	- Removed support for 176LQFP-EP and 257LFBGA package options throughout the document; - Added HSADC-related specification throughout the document; - In <i>Chapter 3 General</i>: - Updated the data in <i>Table 2. Operating requirements</i> and added power sequencing requirements in <i>Section 3.2 Operation Condition</i>; - Added a note on MSC to <i>Table 4. Max functional clock of modules in Section 3.4 Clock Operating Condition</i>; - Updated "V_{HYS}"-related data in <i>Table 5. LVR/LVD/HVD and POR in Section 3.5 LVR, LVD, HVD, and POR Operating Requirements</i>; - Removed the typ. value of "V25" and added the min. and max. values of "C_{OUT}" to <i>Table 10. V25 LDO specification to Section 3.8 PMC Internal LDO</i>. - In <i>Section 4.2 IO DC Specification</i>: - Updated the max. values of "V_{ol_normal}" and "V_{ol_fast}" in <i>Table 13. 3V IO DC specification</i>; - Updated "V_{HYS}", "V_{oh_normal}", "V_{ol_normal}", "V_{oh_fast}", and "V_{ol_fast}"-related data in <i>Table 14. 5V IO DC specification</i>.

Revision	Date	Changes
		- Fixed the typo by changing the description "The FOSC can keep enabled in Standby/Stop mode." to "The FOSC can keep enabled in Stop mode." in <i>Section 5.1 FOSC Specification</i>; - Added <i>Table 25. NVM Program/Erase Time</i> to <i>Section 6.2 NVM Program/Erase Time</i>; - In <i>Chapter 7 Analog</i>: - Added a note on "T_{su}" to <i>Table 25. ADC operation specification</i> in <i>Section 7.1.1 ADC Operation Specification</i>; - In <i>Table 28. CMP specification</i> in <i>Section 7.3 CMP Specification</i>: - Added "V_{DDA}", "V_{SSA}", and "V_{REF_EXT}"-related data; - Changed the min. value of "V_{ref_range}" from "3.3" to "3"; - Removed "100 mV over drive" related data of "T_{prop}"; - Added the max. values of "INL" and "DNL". - Added the subsection <i>Section 7.7 HSADC Specification</i>. - In <i>Chapter 8 Peripherals</i>: - Added the subsection <i>8.1.3 FCSPI Pin Options</i> and reorganized content in <i>Section 8.1 FCSPI Specification</i>; - Fixed a typo in <i>Figure 10. MSC timing</i> in <i>Section 8.2 MSC Specification</i>; - Updated the typ. and max. values of "MII_1" in <i>Table 38. MII/RMII electrical specification</i> in <i>Section 8.3 ENET Specification</i>; - Added the subsection <i>8.4 LU Specification</i>. - Changed the max. value of "t_{dvo}" from "6" to "5" on HS pad in <i>Table 42. Trace block electrical specification</i> in <i>Section 9.2 Trace Block</i>; - In <i>Chapter 10 Package</i>: - Added the thermal simulation data to <i>Section 10.1 Thermal Data</i>; - Added 320LFBGA package outline drawing to <i>Section 10.2 Package Dimension</i>. - Editorial changes in <i>Chapter 10 Package</i> and <i>Chapter 11 Pinout</i>; - Attached the orderable part number list

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